Guidance on implementation of a quality system in blood establishments

World Health Organization

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Preface

Blood transfusion is a key part of modern health care. All blood products must be safe, clinically effective, of appropriate and consistent quality, and available in a timely manner. Every blood establishment (BE) should put in place an effective quality system to ensure the safety of blood products. The quality system should cover all aspects of the BE's activities and ensure traceability, from the recruitment and selection of blood donors to the transfusion of blood and blood products to patients.

For a BE to establish and sustain an effective quality system, it needs to be clearly defined, structured and organized. It must also be tailored to the activities of the BE and comply with national standards and regulatory requirements. As well as management commitment and support, staff motivation and buy-in are critical to success. The BE must therefore be able to mobilize and utilize available resources effectively to implement the quality system. This includes harnessing available tools and programmes, such as those for external quality assessment, surveillance and vigilance, as well as existing quality auditing and inspection programmes, and use of International Biological Reference Materials. World Health Organization (WHO) quality assurance guidelines and the recommendations of other technical documents related to quality assurance of blood products should also be applied.

Ensuring quality is a continuous process, and, therefore, assessment of effectiveness of the quality system should be done continually. The assessment of BEs can include inspection of their implementation of Good Manufacturing Practices; ongoing collection and analysis of data generated from key activities and their use in quality improvement; establishment of haemovigilance through a system of monitoring, reporting and investigation of adverse incidents related to all blood transfusion activities; joining a programme of regular internal and external audits of the quality system; active participation in appropriate external quality assessment schemes to improve laboratory performance, and joining an accreditation programme.

Strategic Objective 3 of the WHO Action framework to advance universal access to safe, effective and quality-assured blood products 2020–2023 aims to achieve functioning and efficiently managed blood services. To meet this objective, High-Level Outcome 2 is to have a functioning quality system in place along the entire blood transfusion chain. Development and application of this guidance is one of the activities that will assist Member States to implement a functioning quality system in their BEs, which will ensure all activities relating to blood products (including the associated substances and in-vitro diagnostic devices required for these activities) are carried out in a quality-focused manner.

Moreover, this guidance will contribute to the thirteenth General Programme of Work, which highlights universal health coverage. This includes appropriate access to affordable and quality-assured medicines, vaccines and health products (including diagnostics and devices, as well as blood and blood products).

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List of abbreviations and acronyms

AABB	Association for the Advancement of Blood & Biotherapies
AfSBT	Africa Society for Blood Transfusion
BE	blood establishment
BRM	WHO Biological Reference Material
CA	corrective action
CE	Conformité Européenne (French for European Conformity)
COMAR	COde d'Indexation des Matériaux de Référence (international database for certified reference materials)
CRM	certified reference material
ECBS	WHO Expert Committee on Biological Standardization
EDQM	European Directorate for the Quality of Medicines and HealthCare
EQA	external quality assessment
EQAS	external quality assessment scheme
FDA	United States Food and Drug Administration
GBT	WHO Global Benchmarking Tool (also, GBT + blood, which is for assessment of the blood regulatory authority)
GMP	Good Manufacturing Practices
HACCP	hazard analysis and critical control points
IQ	installation qualification
ISBT	International Society of Blood Transfusion
IS	(WHO) International Standard(s)
ISO	International Organization for Standardization
IU	international unit
M&E	monitoring and evaluation
MoH	ministry of health
NRA	national regulatory authority
OQ	operational qualification
OHS	occupational health and safety
PA	preventive action
PAHO	Pan American Health Organization
PDCA	plan, do, check, act (from the Deming quality cycle)
PDMP	plasma-derived medicinal product
PIC/S	Pharmaceutical Inspection Co-operation Scheme
PMF	plasma master file

PPE	personal protective equipment
PQ	performance qualification
PQR	product quality review
QI	quality indicator
QMP	WHO quality management project
QA	quality assurance
QC	quality control
RCA	root cause analysis
REMCO	ISO Committee on Reference Materials
SOP	standard operating procedure
SMART	specific, measurable, achievable, realistic, timely
TTIA	transfusion-transmissible infectious agent
VNRD	voluntary non-remunerated donation
WHO	World Health Organization

Glossary

Accreditation	Process by which an independent and authorized agency certifies the quality and competence of an organization or establishment on the basis of certain predefined standards.
Adverse event	Any undesirable or unintended occurrence associated with transfusion or donation. It includes all adverse reactions, incidents, near-misses, errors, unplanned deviations from standard operating procedures and accidents.
Apheresis	Process by which one or more blood components are selectively obtained from a donor by withdrawing whole blood, separating it by centrifugation or filtration into its components, and returning those not required to the donor.
Audit (or assessment or inspection)	A systematic, independent examination performed at defined intervals and at sufficient frequency to determine whether actual activities comply with planned activities, are implemented effectively and achieve objectives. This includes documented review of procedures, records, personnel functions, equipment, materials, facilities and/or vendors to evaluate adherence to written standard operating procedures, standards, relevant laws and regulations. Audits are conducted by professional peers, internal quality system auditors or certification/accreditation body auditors.
Biological product	A product derived from cells, tissues or microorganisms, applicable in the prevention, treatment or cure of a disease or condition. A biological product may be a virus, therapeutic serum, toxin, antitoxin, vaccine, blood, blood component or derivative, allergenic product, protein or analogous product.
Blood cold chain	System for storing and transporting blood and blood products within the correct temperature range and under the correct conditions, from the point of collection from blood donors to the point of transfusion to the patient.
Blood component	A constituent of blood (erythrocytes, leukocytes, platelets, cryoprecipitate, plasma) that can be prepared by various separation methods and can be used either directly for therapeutic purposes or for further processing and manufacturing.
Blood establishment	Any structure, facility or body that is responsible for any aspect of the collection, testing, processing, storage, release or distribution of human blood or blood components (including source plasma) when intended for transfusion or further industrial manufacturing. It encompasses the terms blood bank, blood centre, plasma collection centre, blood service and blood transfusion service. Some hospital-based blood services may additionally be engaged in storage, pre-transfusion testing of patients and issuing of blood components, and function as both blood establishments and hospital blood banks.
Blood grouping	Determination of specified blood groups in donor or recipient blood samples for the purposes of compatibility testing, based on the presence or absence of antibodies and inherited antigenic substances on the surface of red blood cells.

Blood processing	Any step in the preparation of a blood component that is carried out between the collection of blood and the distribution of a blood component by a blood establishment. This includes physical procedures such as separation, centrifugation, modification of temperature gradients etc.
Blood product	Any therapeutic substance derived from human blood, including blood for transfusion (i.e. whole blood and blood components), plasma for fractionation (either separated from whole blood or prepared by apheresis), and plasma-derived medicinal products.
Blood service	The organization responsible for the collection, screening, processing, storage and supply of blood and blood components for therapeutic use. It encompasses the term "blood transfusion service".
Blood system	A blood system encompasses blood regulatory systems, blood supply systems or "services"; blood transfusion systems, including hospital blood banks and clinical transfusion services; related laboratories; and allied industries, including providers of related substances, reagents and medical devices.
Blood transfusion chain	The entire sequence of procedures from the donor to the patient, including donor recruitment and screening, collection of blood or blood component donations, processing and testing of blood and its components, through to their provision and transfusion to patients. Not all steps may be applicable to every blood establishment.
Calibration	The set of operations which establish, under specified conditions, the relationship between values indicated by a measuring instrument or measuring system, or values represented by a material measure, and the corresponding known values of a reference standard.
Change control	A structured method of revising a policy, process or procedure, including hardware or software design, transition planning and revisions to all related documents.
Competence	Ability of an individual to perform a specific task according to procedures.
Compliance (or conformance)	Fulfilment of requirements. Requirements may be defined by customers, practice standards, regulatory agencies or law.
Corrective action	An activity performed to eliminate the cause of a finding of non-compliance or other undesirable situation in order to prevent a recurrence.
Critical	Potentially having an effect on the quality and/or safety of, or coming into contact with, the cells and tissues.
Errors	Errors occur when something that should have been done was either not done or was done incorrectly as a result of poor judgement, wrong decision or wrong action.
External proficiency testing	An external assessment of laboratories to ensure their ability to perform to the level of competence and quality required, through evaluation of participant performance against pre-established criteria by means of interlaboratory comparison.
External quality assessment	External assessment of a laboratory's results from testing exercise material of known, but undisclosed content and objective comparison with the performance of other laboratories that have tested the same material.

Fractionation	(Large-scale) process by which plasma is separated into individual protein fractions that are further purified for medicinal use (variously referred to as plasma derivatives, fractionated plasma products or plasma-derived medicinal products). The term fractionation is usually used to describe a sequence of processes, including plasma protein separation steps (typically precipitation or chromatography), purification steps (typically ion exchange or affinity chromatography), and one or more steps for the inactivation or removal of bloodborne infectious agents (viruses and, possibly, prions).
Gantt chart	A project management tool showing activities (tasks or events) displayed against time. On the left of the chart is a list of the activities and along the top is a suitable timescale. Each activity is represented by a bar; the position and length of the bar reflects the start date, duration and end date of the activity.
Global Benchmarking Tool	A comprehensive set of indicators that constitute the primary means by which the World Health Organization objectively evaluates regulatory systems, as mandated by World Health Assembly resolution WHA67.20 on regulatory system strengthening for medical products.
Good Laboratory Practices	A quality system concerned with the organizational process and the conditions under which non-clinical health and environmental safety studies are planned, performed, monitored, recorded, archived and reported.
Good Manufacturing Practices	All elements in the established practice that will collectively lead to final products or services that consistently meet appropriate specifications and comply with defined regulations. The part of quality assurance that ensures that products are consistently produced and controlled to the quality standards appropriate to their intended use, and as required by the marketing authorization or product specification. Good Manufacturing Practices are concerned with both production and quality control.
Haemovigilance	A set of surveillance procedures covering the entire transfusion chain, from the donation and processing of blood and its components to their provision and transfusion to patients and their follow-up. It includes the monitoring, reporting, investigation and analysis of adverse events related to the donation, processing and transfusion of blood, as well as the development and implementation of recommendations to prevent their occurrence or recurrence.
Hospital blood bank	A hospital unit which stores and issues and may perform compatibility tests on blood and blood components exclusively for use within hospital facilities, including hospital-based transfusion activities. Some hospital-based blood services may additionally be engaged in blood collection, processing and donation testing activities and thereby function as blood establishments.
Incident	Any untoward occurrence associated with an activity or process, such as the collection, testing, processing, storage and distribution of blood and blood products, or with its transfusion or administration.
Internal quality control	Procedures that monitor the day-to-day reproducibility of test results and detect major errors in the analytical process.
Low- and middle-income countries	Resource-constrained countries with a gross national income per capita that is below a value determined each year by the World Bank.
National regulatory authority (NRA)	Legally established bodies that promulgate medicines regulations and enforce them. The NRA is the entity in charge of assuring the quality, safety and efficacy of medicinal products as well as ensuring the relevance and accuracy of product information.

National blood system (or programme)	National system that incorporates the blood regulatory system, the blood supply system and the blood transfusion system.
National blood policy	A formal statement of intent by the national health authority that defines the organizational, financial and legal measures that will be taken to ensure that safe blood and blood products are available for all patients requiring transfusion and are used appropriately and safely. It protects and promotes the health of blood donors, the recipients of blood and blood products, staff, the community and the environment.
National haemovigilance system	A haemovigilance system that is implemented throughout a country, with coordination, data analysis and reporting at the national level.
Near-miss	An error or unplanned deviation from procedure that was detected and corrected before it could have an undesirable or unintended impact.
Near-miss event	An unexpected occurrence that did not adversely affect the outcome but could have resulted in a serious adverse event.
Patient blood management	A patient-centred, systematic, evidence-based approach to improve patient outcomes by managing and preserving a patient's own blood, while promoting patient safety and empowerment.
Plasma-derived medicinal product	A range of medicinal products obtained by the fractionation of human plasma. They are also called plasma derivatives, plasma products or fractionated plasma products.
Plasma for fractionation	Recovered plasma or source plasma used for the production of plasma products.
Preventive action	Action taken to eliminate the cause of a potential non-conformity or other potential undesirable situation.
Procedure	A specific activity that forms the basic unit of a process.
Process	A series of steps or actions that lead to a desired result or output.
Quality	The total set of characteristics of an entity that affect its ability to satisfy stated and implied needs, and the consistent and reliable performance of services or products in conformity with specified requirements. Implied needs include safety and quality attributes of products intended both for therapeutic use and as starting materials for further manufacturing.
Quality control	Part of good practice that is concerned with sampling, specifications and testing, as well as with the organization, documentation and release procedures. The aim is to ensure that materials are not released for use during preparation, and blood and blood components are not released for distribution, until their quality has been judged to be satisfactory and that the necessary and relevant tests have been carried out.
Quality assurance	A part of quality management focused on providing confidence that quality requirements will be met. All the activities from blood collection to distribution performed with the object of ensuring that blood and blood components are of the quality required for their intended use.
Quality indicators	Objective quality measures of key system elements, used to identify potential quality concerns and risks, and to monitor the changes over time.
Quality management system	A management system that directs and controls an organization with respect to quality, and ensures that steps, processes, procedures and policies related to quality activities are being followed.

Quality risk management	A systematic process for the assessment, control, communication and review of risks to the quality of the product throughout the product's life cycle.
Quality system	The organizational structure, responsibilities, policies, processes, procedures and resources established to implement quality requirements.
Safety	The minimization of any health risks to blood donors through the donation process and any health risks to recipients from blood products.
Sentinel event	Unexpected occurrence involving death or serious physical or psychological injury, or the risk thereof.
Sufficiency	A secured supply of blood and blood components sufficient to meet the needs of the country's health-care system.
Specification	A description of the criteria that must be fulfilled in order to achieve the required quality standard.
Standard operating procedures	Written instructions that are adopted and implemented in the blood establishment for the performance of specific procedures in a standardized manner.
Statistical process control	A tool that enables an organization to detect changes in the processes and procedures it carries out by monitoring data collected over a period of time in a standardized fashion.
Supplier (or vendor)	An entity that provides an input material or service.
Testing (in blood establishment laboratories)	Includes transfusion-transmissible infectious agent screening and/or blood grouping.
Traceability	The ability to trace each individual unit of blood or blood component derived thereof from the donor to its final destination, whether this is a recipient, a manufacturer of medicinal products or disposal, and vice versa.
Transfusion-transmissible infection	An infection that is potentially capable of being transmitted by blood transfusion.
Transfusion-transmissible infectious agent	An infectious agent that is potentially capable of being transmitted via blood and components.
Transfusion-transmissible infectious agent screening	Assay systems to detect the presence of markers of infection for identified transmissible infectious agents in donor blood samples.
Validation	The establishment of documented and objective evidence that the predefined requirements for a specific procedure or process can be consistently fulfilled. It includes actions for proving that any operational procedure, process, activity or system leads to the expected results. Validation work is normally performed in advance of implementation according to a defined and approved protocol that describes tests and acceptance criteria.
WHO Model List of Essential Medicines	The WHO Model List of Essential Medicines contains the medications considered to be most effective and safe in meeting the most important needs in a health system. The Model List identifies individual medicines that together could provide safe and effective treatment for most communicable and noncommunicable diseases. This list includes blood and blood components, and plasma-derived medicinal products, namely immunoglobulins and coagulation factors, which are needed to prevent and treat a variety of serious conditions that occur worldwide.

Executive summary

An effective quality system provides a framework within which the activities of the blood establishment (BE) are developed, performed in a quality-focused way and continuously monitored to improve outcomes. This helps to ensure that blood and blood products are safe, adequate to meet demand, effective and produced consistently to the appropriate standards.

The World Health Assembly (through resolution WHA63.12) promotes the establishment of quality systems in BEs as one of the strategies necessary for ensuring the quality, safety and efficacy of blood products. The World Health Organization (WHO) Action framework to advance universal access to safe, effective and quality-assured blood products 2020–2023 has identified a functioning quality system in place across the entire blood transfusion chain as a desired outcome under the strategic objective to promote functioning and efficiently managed blood services.

This document provides guidance on the activities needed to implement a functioning quality system in the BE, with information on the technical resources available. It recognizes that, in order to establish and sustain an effective quality system, BEs must be able to efficiently and effectively mobilize and utilize available resources, and leverage existing tools and programmes.

It describes the benefits and challenges of implementing a quality system in a BE and elaborates on the key elements of an effective quality system, which include:

- organizational management – clearly defined job descriptions, appointment of suitably trained and competent staff, mapping of processes and their critical control points, and the development of a quality culture;
- identification of appropriate national or international standards and guidelines and the use of international reference materials in laboratories;
- development of documentation, including policies, standard operating procedures, a quality manual, forms, records and labels, as well as a system for controlling these documents;
- identification of training needs with a training policy and a plan for the training of staff and the education of other health-care professionals involved in blood transfusion;
- management of processes with procedures for validation, change control, error management, and quality and management reviews;
- monitoring of performance through internal and external audits, participation in external quality assessment schemes and haemovigilance, use of quality indicators, and the continuous collection and analysis of data; and
- the quality cycle and need for continuous quality improvement as ongoing activities within the quality system.

Other important aspects of quality systems are also addressed, such as management commitment and support, management of resources, occupational health and safety, and emergency planning. Key factors for success are highlighted, including leadership involvement, staff buy-in and ownership, and a positive quality culture.

The document explains how to approach implementation in a step-by-step manner, outlining practical considerations and guiding the development of a suitable roadmap. This approach acknowledges that there is no universal process for implementing a quality system in a BE, and that the order and way in which elements of the system are introduced should depend on local priorities, circumstances and available resources. It is nonetheless a dynamic process involving many interconnected parts and a systematic approach will help to identify key steps and potential barriers. A stepwise approach allows smoother implementation and helps to ensure engagement of all key stakeholders.

Quality involves a commitment to continuous improvement within the organization and recognition that no system is ever perfect. There is always potential to improve. Investment in quality and an organizational commitment to its central role in assuring the supply of sufficient and safe blood products in a timely fashion will provide confidence to customers and stakeholders of the BE that their needs will be consistently met. This will foster a collaborative relationship that supports ongoing development of the national health system.

Development of the guidance

This guidance was developed following an initial (virtual) meeting of a selected group of experts in transfusion medicine, WHO Regional Advisers for Blood and Transplantation, and WHO headquarters Blood and other Products of Human Origin staff. The need for guidance and support for Member States on implementing a functioning quality system in their BEs was discussed and established. The group identified the key topic areas, and individual working groups were established to produce the initial text for each of these areas. An editorial group was formed from the working groups and the final guidance was developed by the initial group through an iterative process before being reviewed by a number of WHO-identified external experts.

Declaration of interest by external contributors

Declarations of interest by external contributors acting in their individual capacity were collected, assessed and managed as per WHO policy.[1] The standard WHO form for declaration of interests (DOI) was completed and signed by each expert. The WHO Blood and other Products of Human Origin Team reviewed all the DOI forms before finalizing experts' invitations to participate. No competing interests were declared and therefore no further action was taken.

1 Declarations of interest. In: About WHO. Geneva: World Health Organization; 2021 (https://www.who.int/about/ethics/declarations-of-interest, accessed 5 November 2021).

CHAPTER 1

Introduction

1.1 Purpose and scope of the guidance

Blood transfusion is an essential part of patient care, which has the potential to save lives and improve health when used correctly. However, it also carries a potential risk of adverse reactions, complications and transfusion-related infections. Acknowledging the importance of the provision of safe blood products, the World Health Assembly adopted resolution WHA28.72 in 1975 urging Member States to protect and promote the health of blood donors and of recipients of blood and blood products *(1)*. Establishing a quality system that covered the whole chain from donor recruitment and selection, blood collection and processing, to testing and distribution of blood components was identified as an important step towards achieving this goal.

Despite significant progress in establishing quality systems in many countries, progress remains slow in some parts of the world. In response to this situation, Strategic Objective 3 in the WHO Action framework to advance universal access to safe, effective and quality-assured blood products 2020–2023 aims to have functioning and efficiently managed blood services. One of the high-level outcomes is a functioning quality system in place throughout the entire blood transfusion chain *(2)*. To establish and sustain an effective quality system, blood establishments (BEs) must be able to effectively mobilize and utilize available resources and leverage existing tools and programmes. Doing so will also promote the development of a quality system in BEs that lack one.

This document provides guidance on implementing a functioning quality system in the BE, which will ensure that all activities relating to the provision of a safe blood supply (including the associated substances and medical devices required) are carried out in a quality-focused manner. "BE" refers to any structure, facility or body that is responsible for any aspect of the collection, testing, processing, storage, release or distribution of human blood or blood components (including source plasma) when intended for transfusion or further industrial manufacturing. In the context of the present document, this includes hospital-based blood services that also function as BEs when they are engaged in blood collection, processing, testing, storage and distribution activities. Although they are not specifically included under this guidance, hospital blood banks that are engaged solely in storage, compatibility testing, and release of blood and blood components may also benefit from applying many of the principles described.

The intention of this document is not to provide an exhaustive text on quality and quality systems, or to replace or replicate existing WHO aide-memoires and guides on the topic of quality and quality systems in BEs. Rather it provides guidance and advice on how to develop and implement an effective quality system in a stepwise fashion in a BE, which runs in parallel with the overall development of the blood system within the country. It also provides information on the technical resources available.

1.2 Target audience

This guidance document is intended for:

- BEs;
- policy-makers, health authorities and nongovernmental organizations involved in the oversight of national blood systems and programmes;
- regulatory agencies; and
- international blood experts and consultants, scientific and professional bodies, public health institutions, and education and academic institutions, which are interested in supporting and monitoring quality system activities.

1.3 Relationship to other WHO documents on quality systems

The WHO *Aide-mémoire for national blood programmes: quality systems for blood safety (3)* was produced in 2002 and outlines the key components of an effective quality system within a BE. The *WHO guidelines on good manufacturing practices for blood establishments (4)* was published in 2011 and addresses current and widely accepted principles of Good Manufacturing Practices (GMP) that are relevant to the consistent production of safe and quality-assured blood components in BEs, including donor safety concerns. Guidance was also provided on haemovigilance with the publication of *A guide to establishing a national haemovigilance system* in 2016 *(5)*. The *WHO Assessment criteria for national blood regulatory systems (6)* and WHO Global Benchmarking Tool + Blood (GBT + blood) *(7)* were introduced in 2012 and 2019, respectively, to provide guidance on evaluating national blood regulatory systems.

The information in this guidance document supplements that provided in these other documents.

1.4 Background

The World Health Assembly adopted its first resolution on blood safety in 1975 (WHA28.72), which urged Member States "to enact effective legislation governing the operation of blood services and to take other actions necessary to protect and promote the health of blood donors and of recipients of blood and blood products" *(1)*. Subsequent World Health Assembly resolutions continued to highlight that it is vital to strengthen BEs and ensure the availability, safety and quality of blood products. To this end, blood transfusion services should develop effective quality systems covering the entire transfusion process – from motivation, recruitment, identification and selection of safe blood donors, blood collection, processing, testing and distribution of blood components, to the transfusion of blood and blood products to patients. The quality system should be clearly defined, structured and organized, relevant to the activities of the BE, and comply with national or international standards and regulatory requirements.

The importance of having a quality system was reiterated in 2010 by the World Health Assembly with the adoption of resolution WHA63.12 *(8)* on the availability, safety and quality of blood products. The resolution requests WHO and its Member States to take all necessary steps to establish, implement and support nationally coordinated, efficiently managed and sustainable blood and plasma programmes according to the availability of resources, with the aim of achieving self-sufficiency unless specific circumstances preclude it. The establishment of quality system and GMP guidelines for the processing of whole blood and blood components, and production of plasma-derived medicinal products (PDMPs), with appropriate regulatory control, including the use of diagnostic devices, are identified as key steps towards achieving the overall goal. This includes building human resource capacity through the provision of initial and continuing staff training to ensure the high quality of blood services and blood products. The role of quality assurance (QA) in increasing the global availability of plasma meeting internationally recognized standards and the need to strengthen the technical capacity of national regulatory authorities (NRAs) in assuring appropriate control of blood products was emphasized.

The essential role of safe effective blood and blood products in health systems is further underlined by the inclusion of blood and blood components in the WHO Model List of Essential Medicines since 2013 *(9)*.

1.4.1 Overview of quality and quality systems

Blood is a biological product. Each donation constitutes an individual batch within a production setting, since the contents of the donation will reflect the characteristics of the donor from whom it was collected. Factors that vary between donations include blood groups, the possible presence of transfusion-transmissible infectious agents (TTIAs), and quantitative differences in cell and plasma constituents. The BE is responsible for converting this highly variable starting material into a standardized medicinal product that must be both safe and efficacious. A functioning quality system provides the framework to support this and will also help assure traceability, which is particularly important for biological products.

The *WHO Guidelines on good manufacturing practices for blood establishments (4)* define quality as "the total set of characteristics of an entity that affect its ability to satisfy stated and implied needs, and the consistent and reliable performance of services or products in conformity with specified requirements". Implied needs include the safety and quality attributes of products intended both for therapeutic use and as starting materials for further manufacturing. The consistent application of appropriate standards and processes is necessary to ensure quality. A quality system within a BE directs and controls the organization with respect to quality, and ensures that steps, processes, procedures and policies related to quality activities are being followed. The quality system should cover all aspects of the BE's activities and ensure traceability, from the recruitment and selection of blood donors to the transfusion of blood and blood products to patients. It should also reflect the structure, needs and capabilities of the BE, as well as the needs of the hospitals and patients that it serves.

The key elements of a quality system are shown in Table 1.1. More detailed information on each of these elements is provided in the following chapters of this document.

Table 1.1 Key elements of a quality system

Key element	Requirements
Organizational management	• Clearly defined organizational structure, job descriptions • Management responsibility • Appointment of suitably trained and competent staff • Culture of quality • Mapping of processes and their critical control points
Standards, guidelines and reference materials	• Identification of appropriate national or international standards and guidelines • International reference materials
Documentation	• Quality policy • Quality manual • Standard operating procedures • Labels • Forms and records • A system for controlling documents
Training	• Identification of training needs • A training policy and plan • Training of all staff in quality and quality systems • Education of other health-care professionals involved in blood transfusion • Evaluation of training and its impact • Leveraging available resources and assistance in technical training
Assessment	Part A: Assessment – management of process and improvement • A system for validation of processes, procedures, equipment, reagents • Change control • Error management with associated corrective and preventive actions • Management review Part B: Assessment – monitoring of performance • Internal and external audits • Participation in relevant external quality assessment schemes • Quality indicators • Continuous data collection and analysis • Participation in haemovigilance activities

Successful implementation of a quality system is the responsibility of senior management but also requires participation and commitment from all staff within the organization. This includes staff in many different departments and at all levels, as well as the organization's suppliers and distributors. Participation and commitment is achieved by developing a quality culture which can be described as "an environment in which staff not only follow quality guidelines but also consistently see others taking quality-focused actions, hear others talking about quality, and feel quality all around them" *(10)*. People are human and inevitably mistakes will occur in even the best managed organizations. The quality system should therefore include processes to capture and investigate errors and non-conformances and to use the information collected to identify opportunities for ongoing improvement. This concept should be reinforced during the training of supervisors, and voluntary reporting of adverse events encouraged.

Quality requirements will increase progressively in parallel with the development of the wider organization as it implements more complex systems for the collection, processing, testing and screening of blood, and as it expands the range of blood products that it provides. This principle of continuous improvement, often described as the quality journey, is critical to the successful implementation of a quality system. Regular review of the system is necessary to identify areas for improvement and to provide assurance that the system is performing according to expectations.

1.4.2 Progress in implementation of quality systems in blood establishments

Over the past decades, WHO has provided numerous tools and training activities as well as technical support to enhance the development of national standards and quality systems in many countries. An aide-memoire was produced in 1990 describing the tools provided by WHO that could be used by NRAs and manufacturers in Member States for assuring the quality, safety and efficacy of blood products and related biologicals *(11)*. WHO published the *Guidelines for quality assurance programmes* for blood transfusion services *(12)* in 1993 to provide a comprehensive guide to each component of a QA programme for blood transfusion services.

The WHO Quality Management Project (QMP) for blood transfusion services was initiated in 2000 to assist Member States in establishing national quality systems in blood transfusion services. Under the QMP, capacity-building through structured training courses on quality were organized at regional and country level. The aim was to create greater awareness and knowledge of quality management and to develop a core group of quality managers for blood transfusion services in all Member States. In addition to training materials developed to support the QMP *(13–16)*, the *Aide-memoire for national blood programmes on quality systems for blood safety (3)* was developed as well as WHO publications that provided guidance on various aspects of a quality system *(5,17,18)*.

Support has also been provided to countries for the development of regulatory frameworks, which include the enforcement and implementation of GMP in BEs that collect, process, store and distribute blood products, and regulation of blood safety-related in-vitro devices. The WHO Achilles project *(19)* was implemented from 2002 to 2005 to assist countries in improving the quality of plasma suitable for fractionation, and WHO is working with regulatory authorities and national blood services on the implementation of blood regulatory systems that will help to strengthen quality systems in BEs. The *WHO Recommendations for the production, control and regulation of human plasma for fractionation (20)*, developed in 2005 to guide BEs in implementing appropriate procedures for the production and control of starting plasma material, emphasize the importance of implementing an effective quality system and the key elements of such a system. The *WHO guidelines on good manufacturing practices for blood establishments (4)* provide guidance for both BEs and NRAs on implementing and enforcing GMP principles relevant to the consistent production of safe and quality-assured blood components in BEs. The *Guidelines on management of blood and blood components as essential medicines (21)* also provide a framework for establishing regulatory oversight of blood and blood components for use in transfusion as essential medicines. A list of assessment criteria for national blood regulatory systems is also integrated into the Global Benchmarking Tool (GBT) *(7)*, which is used for evaluating national regulatory systems.

Notwithstanding the progress made in many countries, challenges remain in implementing quality systems in BEs. The Global Database on Blood Safety survey *(22)* noted encouraging progress compared with earlier survey data. However, in many resource-poor countries, inadequacy of blood supply often coexists with inadequate funding, government legislation and regulation. Blood safety also depends on effective TTIA screening, a strong quality system and effective monitoring mechanisms such as haemovigilance systems. In these respects, the survey noted that only 66% of countries reported the existence of specific legislation covering the safety and quality of blood transfusion. Only 59% of countries had a system of regular inspection and licensing by the NRA, and only 33% of countries had an accreditation system for blood transfusion services. Although most countries had national policies for screening for major TTIAs, the effectiveness of TTIA screening can only be guaranteed with well-designed and implemented QA systems. In lower middle-income and low-income countries, only 82.8% and 76.2% of donations, respectively, were screened following basic quality-assured procedures, and only 50% and 68%, respectively, participated in external quality assessment schemes (EQAS) for TTIA testing. Globally only 49% of countries reported having a national haemovigilance system.

The WHO Action framework to advance universal access to safe, effective and quality-assured blood products 2020–2023 *(2)* has set one of the high-level outcomes (high-level outcome 3.2) under Strategic Objective 3 to achieve functioning and efficiently managed blood services as "a functioning quality system in place across the entire blood transfusion chain". Activities to achieve this include the development and dissemination of WHO guidelines and other knowledge products relevant to the key functions of the quality system, and organization of national and regional workshops for national BEs to build capacity for carrying out key functions of the quality system. This guidance document is part of these activities.

CHAPTER 2

Benefits and challenges of implementing quality systems in BEs

Implementation of an effective quality system in a BE will lead to potential benefits, which include improvements in the safety, consistency, availability and quality of blood and blood products provided to patients. It can also improve operational efficiency and cost-effectiveness, and contribute towards the development of better products and processes. However, there may also be myriad challenges in quality system implementation, which must be identified and addressed.

2.1 Potential benefits

2.1.1 Ensures safety, quality and consistency of blood components

An effective quality system based on principles of GMP assures the safety and quality of blood and blood components. Standardization of procedures and practices reduces variation, leading to greater product consistency. The use of appropriate test methods, reagents and test kits ensures the safety of components. Successful implementation of GMP in the BE thus ensures that blood and blood components are consistently produced and controlled according to predefined quality standards appropriate for their intended use. Quality improvement ensures that there is continuous improvement at the BE, while quality risk management proactively identifies potential risks to safety and quality so that they can be controlled.

2.1.2 Improves clinical outcomes

By producing blood and blood components that meet appropriate and defined quality standards, the BE can provide hospitals and patients with products that are consistently safe and efficacious. This helps to improve transfusion outcomes and reduce unnecessary transfusions as well as saving health-care costs. Improved safety of the product results in lower transfusion-associated morbidity or mortality, with fewer adverse transfusion reactions and fewer transfusion-transmitted infections. The overall health and survival of patients requiring blood transfusion is increased.

2.1.3 Improves availability of blood and blood components

Blood donors are a finite and limited resource. Standardized processes, trained competent staff and quality improvements will all help to reduce the number of units discarded because of process deviations and errors. Improving the reliability and consistency of donor screening and selection lowers the prevalence of infection among donors and leads to fewer discards of unsuitable donations. Wastage of donations because products were improperly collected, processed, handled, transported or stored is reduced. Monitoring of blood inventories using quality indicators (QIs) results in a reduction in the number of expired units. With fewer discards and reduced wastage, the potential to produce safe and effective blood and blood components from the donors is maximized and the availability of blood and components improved.

2.1.4 Improves access to PDMPs

Assuring the safety and quality of plasma collected by apheresis or recovered from whole blood enables it to meet the required quality standards for fractionation and to be used as starting material. The ability to fractionate domestic plasma helps the country or region to achieve self-sufficiency in PDMPs. It also supports the implementation of international quality standards. In turn, this enables more PDMPs to be made available and improves global supply and availability.

2.1.5 Ensures donor health and safety, and encourages voluntary non-remunerated donation (VNRD)

Assuring the quality of blood donor selection and blood collection procedures contributes to donor safety and reduces the risk of errors during the collection procedure. Minimizing adverse events in donors and addressing them when they do occur helps to foster donors' confidence in the donation process and helps to promote donor return, donor retention and regular donation. Increased confidence and satisfaction with the blood donation process also encourages VNRD. It improves perceptions of VNRD in the community, thus motivating more healthy people to donate blood, which, in turn, improves blood safety and availability.

In addition, the quality of donor screening has implications for donor health. For example, detection of infections in the donor can enable early diagnosis and treatment. But false-positive screening results can lead to adverse outcomes and even social stigmatization.

2.1.6 Improves productivity and more cost-effective use of resources

Overall productivity in the BE is enhanced when quality is improved. Having an effective quality system in place enables improvements in work processes and safety and enhances work performance. Reduction of in-process variation and errors results in fewer reworks and delays, and less time expended on investigating and managing errors and accidents. Implementing appropriate corrective and preventive actions will also address problems and prevent them from recurring. Routine facility and equipment maintenance reduces the need for repairs and periods of operational downtime, increasing the efficiency and lifespan of the facilities and equipment. Cost reduction and less wastage of blood donations, materials, consumables and human resources leads to more cost-effective use of resources.

2.1.7 Improves staff confidence, participation and morale

By standardizing work policies and processes and providing suitable and appropriate training, staff performance and confidence are improved. This improves job satisfaction and motivation as staff take more pride in their work, which results in increased customer satisfaction and fewer errors. Ensuring a safe work environment and practices enhances staff health, which in turn leads to better productivity and staff morale. This is further strengthened by a strong quality culture in the organization with trained and competent staff who work towards achieving high quality, which also fosters staff participation and commitment. The improved staff confidence and morale will also contribute towards staff retention, reduce staff attrition and improve staff recruitment.

2.1.8 Facilitates inspection and accreditation

Having an effective quality system in place demonstrates the organization's commitment to high standards of product quality, patient safety and compliance with regulatory requirements. This facilitates government oversight and inspection of the BE by regulatory authorities as well as supporting accreditation.

2.1.9 Increases confidence in the blood supply

There is increased confidence and trust in the blood supply when the BE is able to reliably provide safe, adequate and quality-assured blood products to meet clinical needs. The BE achieves greater credibility among its stakeholders, who are more likely to support its activities. There is also less risk of legal action. Positive attitudes towards blood donation and blood transfusion develop, which will boost VNRD.

2.1.10 Enables and supports BE development

BEs will need to develop their capabilities to support progress and advances in the health-care system, whether in provision of a wider range of, or more complex, blood components and PDMPs, or more complex services. Quality systems play an important role in supporting changes of this type and ensuring that change is successfully delivered. Quality systems also provide the platform to identify opportunities for improving current products and processes, as well as to evaluate and address potential pitfalls and risks.

2.1.11 Strengthens national capacity to meet blood needs and supports development of national capabilities

At the national level, a quality system promotes the efficiency and effectiveness of the blood system as well as greater availability of safe and quality-assured blood products and reduction of blood production costs. It also allows the incorporation of national and international regulatory frameworks, facilitating the exchange of blood products locally and regionally and thus improving equitable access to transfusion therapy. In addition, security of the blood supply can be assured.

2.2 Potential challenges and risks

2.2.1 Lack of political commitment and support, and failure to appreciate the societal cost of insufficient supply of safe and quality-assured blood products

Lack of political will among the decision-makers to acknowledge the importance of blood transfusion and the role of BEs in the provision of an adequate supply of safe and quality-assured blood products is a major challenge. This lack of political commitment and support will potentially lead to a lack of investment in the organization and governance of BEs and a lack of the resources required to assure the adequacy, safety and quality of blood products. Transfusion medicine is often not recognized as a specialty in medicine and ensuring an adequate number of competent staff remains a challenge.

2.2.2 Absence of a nationally coordinated blood service, national policies and standards

An efficient blood service should be well organized and nationally coordinated with national policies, strategies and standards. Blood services that are not well organized and appropriately consolidated rely on multiple small-volume BEs, which limits the ability to achieve compliance with accepted standards. Without economies of scale, it becomes difficult to maintain an adequate number of competent staff. Fragmentation also makes proper blood donor management expensive and operationally difficult.

2.2.3 Insufficient or absent regulatory frameworks and oversight

Although quality systems in BEs should ideally comply with requirements set by the regulatory authorities, blood regulation is not yet developed in many countries. Regulatory frameworks are either absent or largely inadequate to regulate the entire vein-to-vein system or the medical devices used in the manufacture of blood and blood components. In many instances, blood regulatory authorities are nonexistent or have not yet started regulating the blood sector. There may be no authorized and empowered competent regulatory authority to define and enforce the standards for blood collection and processing, including GMP. This could be because, historically, BEs have been regulating themselves. It might also be due to lack of recognition by the regulatory authorities that whole blood, blood components and PDMPs are therapeutic products included in the WHO Model List of Essential Medicines.

Technological advances in the collection and processing of blood for transfusion have changed the blood preparation model from simply a "medical service" to a complex process of "product manufacturing" under defined standards. Laws, regulations and the adoption of quality processes consistent with this concept have yet to be developed in some regions.

2.2.4 Unreliable supply management systems

The flow and storage of materials required to provide a reliable supply of safe and quality-assured blood products to meet clinical needs remains a challenge. Lack or weakness in evaluation of suppliers may result in irregular supply, suboptimal handling of materials and incorrect or poor-quality materials being provided. The logistics of moving supplies from their place of origin to the BEs may be complicated and requires a high level of oversight and expertise, further reinforcing the need for quality management in BEs. Furthermore, equipment calibration services are often unfamiliar with the requirements of blood services.

2.2.5 Absence of, or lack of access to quality control programmes and EQAS

Limited knowledge of its importance results in the absence of quality control (QC) to assure the accuracy and reliability of test results or consistent product yields. This may be compounded by the lack of access to controls for reagents and test kits leading to inadequacy or absence of controls for reagents and assays and related laboratory practices. Lack of access to EQAS results in failure to identify or recognize the magnitude of product quality deviations from acceptable standards.

2.2.6 Limited awareness and knowledge of specific blood products or the concepts of patient blood management

Lack of information, knowledge and awareness about appropriate blood component therapy and patient blood management among clinicians may arise from insufficient investment in training of medical staff and transfusion medicine specialists. Improper or inappropriate use of blood and blood components results in patients not receiving optimal blood transfusion therapy. This may also affect financial coverage for blood component transfusions, which also limits access. Similarly, lack of knowledge about patient blood management may lead to inefficient use of blood and blood components and hamper the adoption of alternative therapies and technologies that help to minimize blood transfusions and conserve the blood supply.

2.2.7 Limited financial resources

Implementation of a quality system often requires the investment of additional resources, especially during the initial stage. If a BE decides to establish a quality system, it will need to provide sufficient budget for any extra staff employed specifically for quality management, and new programmes for training and audits, among others. This could present a major challenge for BEs in resource-limited countries, especially for those unable to access external funding support or operating on a cost-recovery basis *(22)*. In such situations, a stepwise approach to implementing the quality system should be considered.

2.2.8 Potential for adverse effects on staff morale and confidence

Although the presence of a functioning and effective quality system can have a positive effect on staff morale and confidence, the initial impact may be adverse in systems that were previously ineffective or non-existent. As the quality system becomes more reliable and systematic observation exposes hidden problems, the pressure on staff and the organization increases. In contrast, in organizations with no quality system, these problems are not detected and thus there is no pressure. This may result in staff opposition to quality system-related activities. Such opposition should be anticipated so that mitigation measures can be taken as part of quality system development and implementation.

CHAPTER 3

Elements of an effective quality system

3.1 Key elements

Whereas the specific elements of a quality system may vary depending on national laws and the need to adapt to different blood transfusion chain practices, the following are the key elements of an effective quality system *(3)*:

- organizational management;
- standards, guidelines and reference materials;
- documentation;
- training; and
- assessment.

BEs should ensure that their implementation plans include activities to strengthen these elements. Chapters 4 to 9 provide detailed guidance on these activities.

There is some variation in terminology in quality standards and guidelines produced by different organizations *(23–26)*, and other quality system aspects besides the elements mentioned above may be emphasized, such as occupational health and safety (OHS) and waste management. BEs should also consider quality system components such as management commitment, and resources including staff, premises, equipment, consumables and budget when planning quality system implementation. The quality requirements in these areas are briefly described below. More information is provided in the *WHO guidelines on good manufacturing practices for blood establishments (4)*.

3.2 Management commitment and support

Attainment of the quality policy and objectives of the BE should be the responsibility of the senior management, with participation and commitment from all staff throughout the entire BE *(4)*. Once the decision to establish a quality system has been made, the organization must be committed and prepared to support the decision, and invest the necessary human, infrastructural, material and financial resources. The BE should have a defined management team in place with the responsibility and authority to establish or make changes to the quality system. Senior management should participate in management reviews of the quality system and ensure that the activities of the BE comply with all relevant standards, laws and regulations *(25)*.

3.3 Resources

3.3.1 Staff

The most important resource of an organization is sufficient, competent and motivated personnel. Attention should be paid to the distinction between personnel and positions. Positions should be set up in accordance with the needs of the blood transfusion chain processes, and job descriptions should accurately set out the required qualifications and skills, tasks, roles, responsibilities and authorities of positions. Only suitably qualified individuals should be appointed to a position. All staff should be given initial and continual training to assure the quality and safety of blood and blood components. The competency of staff should be evaluated upon completion of each initial training topic and on a regular, recurring basis.

3.3.2 Premises

The premises of a BE should enable a safe and efficient workflow that reduces risks of errors and contamination throughout the whole manufacturing chain from donor selection to storage and distribution. Premises should also ensure security and confidentiality of donor information, and the design should allow the safe operation of equipment and effective cleaning of all surfaces. Important aspects that should be addressed include laboratory safety and hygiene, prevention of contamination, access restriction and dress codes. Attention should be paid to the flow of people, goods and waste within the premises and crossing of lines should be avoided. In terms of building structure, area and other detailed requirements, the BE should comply with the provisions of national guidelines, if these exist, as well as GMP requirements. In the absence of national guidelines, guidelines from WHO may be followed *(4, 27)*.

3.3.3 Equipment

Numerous different items of equipment are used in a BE, many of which can potentially affect the quality and safety of a product or process if they do not perform correctly. Equipment management is therefore a critical component of a quality system and includes evaluating the equipment for suitability in the BE, and then qualifying and validating it to demonstrate correct installation and acceptable performance before it is put into routine use. Equipment performance should continue to be monitored regularly, and a maintenance and service programme must be in place.

Adhering to maintenance and service programmes can be challenging in resource-limited countries due to the costs of contracting external service providers. However, this is essential to prevent unnoticed deterioration in equipment performance that can impair product quality or safety. Where resources are limited, skilled technical support or spare parts for specialized equipment may also be lacking or difficult to access, which hampers the ability of the BE to properly maintain and repair the equipment. This should always be a key consideration when the BE is evaluating equipment for purchase. It is also a useful practice for the expected estimated costs of regular maintenance visits and frequently needed spare parts for the first few years to be provided during evaluation so that the cost of maintaining that equipment can be considered. Setting up national or regional networks of BEs which can cooperate or collaborate with each other on these aspects may be helpful.

In many countries, equipment may be leased rather than purchased. This may be a stand-alone leasing option, or as part of a high-volume purchase of consumables or test kits. It is important to know whether the lease agreements include a full maintenance plan. Where maintenance is part of the plan, the costs are usually included in the price of the package.

3.3.4 Reagents and consumables

The management of reagents and consumables in a BE is often challenging, especially for those in resource-limited countries. However, proper procurement and inventory practices can result in cost savings as well as ensuring that reagents and consumables are available when needed. A sound system of procurement, storage and inventory management should be implemented in the BE to ensure that all reagents and consumables are of good quality, and that they are used and stored in a manner that preserves their integrity and reliability. The inventory should be managed following the principle of first-in, first-out, which means that the oldest items, or those closest to expiry, should be the first to be used.

Supplier evaluation and supply verification are among the strategies for ensuring that the inputs to a process meet specifications. The performance of reagents should always be verified after delivery and before routine use. Quality certificates submitted by the manufacturer with each delivery of reagents should be checked. Batch validation in the BE should be considered. Reagents that do not fulfil the acceptance criteria set out by the BE should be returned to the supplier and, where appropriate, compensation or exchange of goods arranged.

3.3.5 Budget

Insufficient budget and budget shortfalls are a major risk for BEs with limited financial resources. To manage this risk, the BE should take steps early on to identify and secure sufficient financial resources to perform, verify and manage its critical quality system activities. There are two types of BEs, those that are autonomous and self-funded through cost-recovery, insurance and/or donor funding and those whose budgets are funded primarily by the government. In the latter case, the primary responsibility of senior management is to develop a budget proposal that supports compliance with the standards, and to advocate for funding at the appropriate ministry, parliament or legislature. In either case, the BE should be responsible for keeping within its budget. A transparent and sound system of financial management should be in place that includes clear accountability for expenditure. All budgets and financing must be accounted for internally as well as undergoing independent audit.

BEs are encouraged to develop longer term financial plans that project costs and revenues for several years to maintain financial sustainability of their operations. Budgets should be developed to ensure ongoing operation with a steady supply of all capital, consumable and disposable items, including buildings, leases, equipment, maintenance, blood bags, reagents, personnel and other items or services needed to ensure that adequate supplies of safe blood and blood products are available.

3.4 Occupational health and safety (OHS)

Ensuring that BEs are healthy and safe workplaces helps to improve the quality and safety of donor and patient care, retention of staff and environmental sustainability. OHS programmes aim to prevent diseases and injuries arising out of, linked with, or occurring in the course of work. An effective OHS programme promotes protection and investment in the BE's most important asset – its staff – and may result in improved productivity, decreases in absenteeism, work premises being maintained at a higher and more comfortable standard, and staff satisfaction.

OHS programmes should be formulated, implemented, monitored, evaluated and periodically reviewed by management in consultation with staff (Box 3.1). They should also align with national laws and regulations. Key elements of an OHS programme are outlined in the *WHO guide for the development and implementation of occupational health and safety programmes for health workers (28)*. The BE should provide continuing (or periodic) education and training to staff on fundamental OHS as well as specialized training appropriate to the nature of the work being carried out.

Box 3.1 Practical tips for a BE to promote health and safety in the workplace

- Develop a safety culture supported by management.
- Appoint a staff member with overall responsibility for the OHS programme.
- Consider establishing a safety committee with a representative from each work area.
- Hold regular meetings and issue minutes detailing follow-up action.
- Use posters and graphs to display information on health and safety.
- Incorporate responsibilities for health and safety into each job description.
- Conduct regular safety checks and inspections against a checklist which is periodically updated.
- Alternatively, incorporate safety checks into the internal quality audits.
- Arrange periodic external assessment by an independent professional body.
- Ensure that all personal protective equipment issued is fit for purpose and correctly used.
- Report and document accidents, injuries and near-misses.

3.5 Emergency and contingency planning

Although the nature of emergencies and disasters may differ, the manner in which the BE responds to the events should be similar. Emergency management policies and practices should be built in as part of the quality system. Detailed information on managing the blood supply during emergencies, including infectious disease outbreaks, is provided in the WHO *Guidance on ensuring a sufficient supply of safe blood and blood components during emergencies (29)* and WHO guidance on *Protecting the blood supply during infectious disease outbreaks (30)*.

One important aspect that is often forgotten during emergencies is risk communication, which addresses not only the nature of the risk, but also the concerns, opinions or reactions of individuals and the community to the information provided. This is important because it provides the opportunity to communicate risks in a planned way that is sensitive to the community and incorporates its members into the process of risk management. Further information and advice about risk communication is provided in the WHO publications on *Communicating risk in public health emergencies (31)* and *Management of national blood programmes (32)*.

3.5.1 Risk assessment, mitigation and preparedness

Risk assessment involves comprehensive evaluation of the risk and its impact, which can be operational, financial or reputational. The acceptable levels of risks that the BE is willing to assume should be considered. Defining the level of risk requires estimating the probability of occurrence and the severity of the consequences *(33)*.

Complex mathematical formulae can be used to estimate the likelihood and probability of risks and their consequences. However, a good place to start would be the simple categorization of probability into frequent, occasional or rare, and classifying severity and/or impact as high, moderate or low. The WHO publication on *Management of national blood programmes* provides information on how to categorize risks in BEs *(32)*.

Risk mitigation considers what measures are in place or should be taken to mitigate the impact of the risks that have been identified and defined. It is an ongoing process. Risk mitigation for blood safety can be classified as donor-related, recipient-related, personnel-related or process-related. Some examples of donor-related risks and the corresponding mitigation strategies are shown in Table 3.1.

Table 3.1 Examples of donor-related risks and mitigation strategies

Risks	Mitigation strategies
Local complications of blood donation Haematoma, arterial and nerve injury	• Good phlebotomy techniques • Vein detectors • Bandages and cold compresses • Adequate post-donation information for donors
Systemic complications Vasovagal reactions, seizures	• Pre-donation information for donors • Avoiding the sight of blood • Pre-donation hydration

Risk preparedness involves the development of plans for how to identify and respond to risks such as unexpected disasters. In most of these situations, actions need to be developed and coordinated both within the organization and with other health sector partners. Specific timelines are important to ensure timely response. The *WHO guidance for contingency planning (34)* provides advice on the actions that should be included as part of risk preparedness.

3.5.2 Contingency planning

Contingency planning manages the "what ifs". It provides the opportunity to identify potential problems in advance and decide how to avoid their occurrence and how to manage them if they do occur *(32)*.

The first step in contingency planning is to identify and rank the risks, hazards and specific scenarios that have been pointed out during risk assessment. Contingency plans should be developed for each type of threat. Working on realistic scenarios provides the basis of accurate planning and helps to identify a potential crisis and the necessary response. Most contingency plans require multisectoral and interagency coordination to ensure maximum synergy of the preparedness and response actions and the best use of resources *(35)*.

A preliminary response plan should include:

- determination of the needs and scope of the response;
- identifying the needs of the affected population; and
- identifying and securing necessary resources.

Regular review is crucial: plans need to be constantly reviewed, tested, updated and improved. This is the best way for the BE to ensure it has the management capabilities to respond in an emergency. Exercises are an integral part of contingency planning and the *WHO guidance for contingency planning (34)* addresses selection, planning and costing of exercises as well as providing an exercise planning tool.

3.5.3 Business continuity planning

Business continuity planning ensures that the BE maintains its ability to provide a sufficient blood supply and critical services, by taking proactive steps to prevent or minimize disruptions and developing plans for how it will recover and restore critical functions. The COVID-19 pandemic increased awareness in BEs of the need to reassess their plans and preparations for unexpected events that could disrupt their operations. For example, supply chain disruptions are now common, and it is important to regularly review and, where necessary, diversify the supply chain to reduce the impact of disruptions on the BE's operations and the blood supply *(29,36,37)*. Continuity plans are also an essential element of blood bank computer systems.

Continuity plans, unlike disaster plans, are not implemented at the time of the event; rather they are activities integrated into the day-to-day operations to ensure consistency, back-up and recoverability. Continuity planning and preparation should be an essential part of daily activities and quality processes.

CHAPTER 4

Organizational management

4.1 Organizational structure and job descriptions

Quality is the responsibility of everyone in the organization. The implementation of a quality system affects all levels of the BE and success requires support from every department and individual. Everyone needs to be involved from the start.

To do their job well, staff must be clear about what is expected of them and how they can develop their job capabilities and competencies to improve further. Tasks need to be clearly defined and staff need to be given the necessary level of authority to carry out these tasks. Staff have a continuing obligation to take responsibility for their work and to be personally accountable for the performance of assigned duties. It is therefore important that the roles and responsibilities of each position in the organization are clearly established, documented and communicated to the staff. Job requirements should be defined and documented in job descriptions.

There should also be well-defined career pathways for staff progression, accompanied by appropriate training programmes to support staff development. In determining and establishing roles and responsibilities, national regulatory requirements should also be met.

4.1.1 Organizational structure

The BE should identify, in an organizational chart that is kept up to date, the hierarchical structure of the establishment, the job titles and areas of responsibility of all key staff. The chart should clearly delineate staff roles and responsibilities, and lines of authority and reporting.

The person with overall responsibility for management of the quality system (quality manager or director) should be clearly identified. BEs in the process of developing and implementing a quality system should assess their current structure and make the necessary changes.

The organizational structure of BEs will vary from country to country and it is necessary to consider all interrelationships within and between departments and any external bodies such as government authorities. Where the *WHO guidelines on centralization of blood donation processing and testing (38)* are followed, relevant roles, responsibilities and interactions should be reflected in the BE's organizational chart.

4.1.2 Job descriptions

Staff should be qualified to perform their tasks. Qualifications, expertise and competence of staff are key prerequisites for job performance and should be clearly documented in job descriptions that are controlled and kept up to date (Box 4.1). Special knowledge and skills needed, such as proficiency in other languages or information technology, and any hazardous aspects requiring specific safety training, such as biohazard and radiation safety, should also be included in the job requirements. Job descriptions should also set out quality-related responsibilities.

Job descriptions should be used to determine the suitability of potential candidates when searching for and hiring new employees. They can also be used to assess the job performance and competency of staff, and to identify training needs. Further information on the role of job descriptions in training is provided in Chapter 7.

It is the responsibility of senior management to ensure that job responsibilities including decision-making authority are clear and unambiguous, that they are delegated and disseminated throughout the organization hierarchy, and that they are reflected in the job descriptions. In BEs with a limited number of personnel, staff may have multiple responsibilities and perform tasks in several work areas. The job descriptions should reflect these diverse requirements and the staff should be trained in all relevant tasks.

Box 4.1 Helpful tips for creating organization charts and job descriptions

- Show relationships with any external bodies such as government authorities on the organization chart.
- In addition to the main organization chart, draw up individual organization charts for each work area.
- Consider using colour for different levels in the organization chart.
- Use job titles not people's names.
- Nominate deputies to cover for key posts in either the organization chart or job descriptions.
- Always ensure organization charts and job descriptions match: when updating one, update the other as well.
- Standardize the format for job descriptions throughout the BE by using a template.
- Keep job descriptions concise; do not duplicate information from standard operating procedures.
- Include responsibility for the quality system as well as quality-related tasks in all job descriptions.
- Job descriptions should be signed by the incumbent and a manager at the next highest level of authority, and possibly at the level above that.
- Remove superseded organization charts and job descriptions from work areas; archive one copy.

4.2 Appointment of trained and competent staff for the performance of quality work

Senior management should be responsible for determining the goals for the quality system, making available the resources to achieve these goals, and promoting a culture of quality in the BE. The responsibility for developing and implementing the quality system to meet these goals is usually delegated to the quality department.

Critical to the implementation of the quality system are quality staff who demonstrate competence and integrity and who recognize the importance of their work and contributions to the overall success of the blood service. Quality staff should develop stewardship of the quality system by providing guidance and oversight of how the system operates.

4.2.1 Quality manager

A designated individual, with training, competence and experience in quality management, should be appointed with the authority and overall responsibility for quality within the BE. This individual should be responsible for the implementation and maintenance of the quality system. The person must report directly to a technically or medically qualified individual at the top management level, who has no direct managerial role in collection, processing, testing or distribution of blood.

The quality manager (QM) assembles and leads a quality department, ensures staff are adequately trained, facilitates reporting of QIs and incidents, and determines which quality improvement actions should be given priority. The QM reports to senior management on the safety and quality of products and services as well as any potential risks.

4.2.2 Quality staff

The availability of experienced, knowledgeable and skilled quality staff facilitates quality system implementation and helps improve quality outcomes.

The quality staff should have the appropriate education and competencies needed for their roles, and be provided with sufficient training and continuing educational development to keep their knowledge and expertise up to date in areas such as:

- process maps;
- policies, procedures, work instructions and records;
- regulatory requirements;
- incident reporting and process improvement; and
- assessment, auditing and corrective actions.

Additional details about key quality staff, their roles and responsibilities and minimum qualifications can be found in the *WHO guidelines on good manufacturing practices for blood establishments (4)*.

4.2.3 Quality teams

Quality teams may also be established to support the quality department. They usually consist of representatives from different work functions who provide expertise and guidance related to their own processes. The best and most experienced staff with expertise in their jobs should be selected for the quality team, and their experience used to implement and improve the quality system. Members of the team continue to report to their line managers in their routine duties but respond to the QM regarding quality activities undertaken as part of the quality team.

The *WHO Quality Toolkit (39)* recognizes that a quality system requires working together through various interrelated aspects and it is therefore important to ensure the quality department and quality teams focus on promoting the following:

- sharing compatible values, vision and purpose;
- interactions based on compassion, respect and dignity;
- widespread, active and inclusive participation;

- shared decision-making processes; and
- equitable and dynamic flow of power, leadership and resources.

If the BE does not have all the competencies needed, it should seek external support. An experienced consultant with the appropriate knowledge and skills can help the BE to better understand the requirements and how to meet them.

4.3 Quality leadership

Effective implementation of a quality system requires more than the development of processes and procedures. Success depends on the BE staff understanding why quality is important and their acceptance of the need for control. *EU Directive 2005/62/EC (40)* states that "Quality should be recognized as being the responsibility of all persons involved in the processes of the blood establishment, with management ensuring a systematic approach towards quality and the implementation and maintenance of a Quality System". The *Good practice guidelines* developed by the European Directorate for the Quality of Medicines & HealthCare (EDQM) *(41)* state that "attainment of this quality objective is the responsibility of executive management. It requires the participation and commitment of staff in many different departments and at all levels within the organization by the organization's suppliers and by its distributors".

Senior managers can consider adopting strategies to demonstrate commitment to quality within the organization and signal to staff that quality is seen as essential (Box 4.2). It is important to adhere to the principle that everyone within the organization is subject to the same rules and requirements when it comes to quality. For example, everyone should comply with privacy obligations and follow all procedures and requirements pertaining to personal protective equipment (PPE) when actively working in donor areas and the production environment, regardless of their position. The principle should be "quality is my responsibility as well as yours". Failure to respect this principle will undermine the quality system.

Box 4.2 Helpful tips on demonstrating management commitment to quality

- Build quality into the way that the organization is structured and directed.
- Ensure that quality principles and requirements are built into projects and initiatives.
- Be seen to adopt and follow quality requirements in day-to-day work and in interactions with staff.
- Actively participate in the quality review process.
- Manage all accidents, incidents and errors.
- Regularly communicate results of quality indicators to staff.

4.4 Developing a quality culture

Every organization has a culture. This is simply a way of saying how an organization expresses itself internally and externally. The culture is driven by values, whether on purpose or by default. The *Harvard business review (10)* defined a "true culture of quality" as "an environment in which employees not only follow quality guidelines but also consistently see others taking quality-focused actions, hear others talking about quality, and feel quality all around them". This involves creating a culture of trust, participation and communication in which quality goals are underpinned by employee satisfaction. There is considerable evidence of the benefits of achieving a true culture of quality within an organization (Box 4.3).

Box 4.3 Benefits of a quality culture

- more engaged and effective employees;
- shared stewardship of the quality system;
- improved productivity and efficiency;
- reduction in errors and rework;
- improved staff retention and reduced sickness and absenteeism;
- improved responsiveness to change; and
- increased ability to recruit good employees.

Organizations with positive quality cultures will see improvements in productivity as their systems are reviewed and refined to support more value-added work, and as processes are analysed to identify and then eliminate non-value-added activities. People will work "smarter" not harder, and consequently will be more engaged and committed to the organization. Involvement of frontline staff in the process fosters "ownership" of the changes and increases capability within the organization. Frontline staff have a deep understanding of the work that they perform and are therefore best positioned to identify ways in which the processes that they use might be improved. Using this depth of knowledge is not only a sensible way to improve efficiency and effectiveness but also improves the staff's sense of self-worth and commitment to the organization. This creates a win–win situation for both staff and the wider organization.

The Malcolm Baldrige quality criteria *(42)* are five essential building blocks of a culture of quality as set out in Table 4.1.

Table 4.1 The five essential building blocks of a quality culture

Criterion	What does it mean?
Open communications	• Staff need access to all the pertinent information to do their jobs. They also need to know when things are going well and when they are not. • For example, senior management should openly share strategic goals and plans. This will provide blood establishment (BE) staff with the direction they need, and more importantly, identify how they can improve. Information that is not specifically related to the job tasks may be equally important, e.g. the cost of production, cost of energy, cost of staff, earnings, losses, risk factors, etc.
Freedom from fear	• Staff need to feel free to share ideas and suggestions without risk of retaliation. • An essential element of a quality culture is that staff feel free to report failures without blame or reprisals. • It is important for everyone in the organization to move away from a "blame the person" mentality to a "blame the process and let's fix it" approach to problems and improvement. • The philosophy should be "there are no successes or failures, just learning experiences".
Data-based decisions	• Decisions should be based on data that are appropriate, accurate, reliable and relevant. • Staff need access to knowledge, facts and data to understand the BE's work processes and environment, including quality indicator results. This helps them to understand the limits of their work, the critical control points and how to manage their tasks and responsibilities. • Training is needed to understand the meaning of data and how this information can best be used to make objective decisions. • Quality teams should regularly gather and analyse data to improve work processes and reduce errors, and every employee needs to feel and be part of these activities.

Table 4.1 *continued*

Criterion	What does it mean?
Trust	• Trust works in all directions and requires open, transparent and consistent communications. • Trust is based on honesty, fairness, respect for the opinion of others and integrity. • BE staff need to be able to trust and rely on each other and on the organization. A mentality of "we're all in this together" (the organization, its suppliers and its customers) is important in building and maintaining trust.
Empowerment	• Commitment involves staff assuming responsibility for initiating and creating success. Commitment, however, should be earned by making staff feel valued and heard. An important way to encourage this is by creating a culture where people listen to one another. • Commitment also goes hand in hand with empowerment, ensuring that staff have the knowledge, skills and authority to act within their prescribed roles. It helps to ensure that their goals align with those of the BE and that they participate in stewardship of the quality system. • Commitment requires trust. Once staff are clear about their roles and goals, the management needs to ensure that they have the authority and freedom to do their jobs without the need for constant supervision. Committed staff respond appropriately to the needs and opportunities they face every day in the BE, ensuring safe products, safe operations, customer satisfaction and continuous improvement. • Establishing an empowered and committed workforce will take time. Staff will need time to adapt to a culture that promotes self-determination. As the quality system evolves, so will the staff and teams. This process can be facilitated by ensuring that BE staff can have a say on what they do and how they can do it better.

4.5 Mapping of processes

The work of the BE consists of interrelated processes. Understanding how results are produced by each process, and how they contribute to the system as a whole, is a fundamental step towards improving effectiveness and efficiency. Mapping enables the BE to understand the important characteristics and activities of each process and to ensure a common understanding of the process by:

- helping to define process boundaries, process ownership, process responsibilities and process metrics (both efficiency and effectiveness measures);
- clarifying tasks and the sequence of activities as well as roles, responsibilities, authority, accountability and process interdependencies; and
- identifying bottlenecks, repetition and delays.

Flowcharts and process maps can be used to identify critical control points for each process using a methodology such as hazard analysis and critical control point (HACCP). A critical control point is the point or step where failure to comply with standard operating procedures (SOPs) could cause harm to donors, patients, staff or the organization itself. It is a point where control can be effectively applied to prevent harm. This allows the BE to put in place adequate and necessary control strategies at steps where it is essential to prevent, eliminate or reduce a risk to an acceptable level.

Additional information on process mapping and the application of HACCP can be found in the WHO guidance document *Application of Hazard Analysis and Critical Control Point (HACCP) methodology to pharmaceuticals (43)*.

CHAPTER 5

Standards, guidelines and reference materials

Standards and guidelines are important for ensuring that blood products are safe and meet quality requirements, that processes are controlled and that the activities of the BE comply with regulations.

It is important to differentiate between standards and guidelines.

Standards for BEs are defined as:

- imperative written statements, stated as positives rather than negatives;
- required goals, not methods;
- scientifically based, clinically sound medical practice, unambiguous requirements, principles associated with GMP, QA and applicable regulations that can provide the basis for accreditation or certification;
- internationally agreed by experts in the field.

Guidelines for BEs are defined as:

- any information product developed by an internationally recognized expert body that contains recommendations for clinical practice or public health policy.

5.1 Standards

The focus of BE standards is to advance quality and safety measures. BE standards combine internationally accepted quality system requirements with relevant technical requirements. Standards also serve as the basis for accreditation. A standard usually contains the word "shall", which indicates that fulfilment is mandatory and that failure to meet the requirement will constitute a non-conformance.

International Organization for Standardization (ISO) standards *(44)* are developed through the consensus of globally established experts, regulators and governments and provide a strong management basis that can be applied in the development of national and international regulation. Standards that are specific to BE activities are available from organizations such as the Association for the Advancement of Blood and Biotherapies (AABB) *(25)* and the EDQM *(41)*, and are used widely by many BEs worldwide. Such standards are often used for purposes of accreditation or licensing of facilities involved in the collection, testing, processing, storage and distribution of blood and blood products, including PDMPs.

The AABB has provided standards formats for use by public health systems in African, Caribbean and Latin American regions to develop region-specific and national standards. These standards formats, which incorporate blood banking terminology and are compatible with the universally accepted ISO9000 standards, were developed to be used as the foundation for standard-setting and accreditation programmes in any region.

Several regional and national organizations have also developed their own standards, which they have kept up to date. For example, the Africa Society for Blood Transfusion (AfSBT) has a set of standards that provide a benchmark for accreditation in developing countries *(26)*. These standards are used in a stepwise programme towards accreditation, which recognizes that BEs operate at different levels of development. The standards progress from basic quality and operational requirements to an intermediate stage and then to full accreditation at international level. The AfSBT standards have been accredited by the International Society for Quality in Health Care External Evaluation Association (ISQua EEA). WHO regional offices have also developed regional standards, such as the Caribbean Regional Standards for Blood Banks and Transfusion Services (45) developed by the Pan American Health Organization (PAHO).

In the absence of a national standard, BEs may wish to develop their own standards tailored to the local context. This could be done by adapting existing standards from other sources *(46–49)* and BEs are advised to research what is available. When adapting standards from other sources, BEs should bear in mind that some standards may not be appropriate to their local situation. For example, many standards for whole-blood-derived blood components are based on whole blood collection volumes of 450 ± 50 mL. BEs that process whole blood units with standard volumes less than 450 mL will need to understand how these requirements are derived and adjust them accordingly. Another example concerns the mandatory tests for TTIAs, which should be based on the local epidemiological situation rather than adopted in their entirety from countries or regions with different patterns of prevalence or spread *(50)*. When adapting standards, any changes should be well-justified, and any risks associated with these changes carefully considered. Approval by national authorities or accreditation by a recognized body should also be sought if the BE is developing its own standards.

5.2 Guidelines

Guidelines and recommendations are statements designed to help end-users make informed decisions on whether, when and how to undertake specific actions. Examples of actions include clinical interventions, diagnostic tests or public health measures carried out with the aim of achieving the best possible individual or collective health outcomes. Guidelines are usually subject to a rigorous QA process that helps to ensure that each and every published guideline is trustworthy, impactful and meets the highest international standards. The development of global guidelines ensuring the appropriate use of evidence represents one of the core functions of WHO. WHO's Guidelines Review Committee ensures that WHO guidelines are of a high methodological quality and are developed through a transparent, evidence-based decision-making process.

Another form of guidelines are "guidance documents". A guidance document is a statement of general applicability issued by a recognized authority such as WHO or an NRA to inform the public of its policies or legal interpretations. For example, in the United States, the Food and Drug Administration (FDA)'s Guidance documents *(51)* provide the agency's interpretation and current thinking on a regulatory issue and usually discuss specific products or issues that relate to the design, production, labelling, promotion, manufacturing and testing of regulated products.

5.3 Accessing standards, guidelines and guidance documents

Written standards, guidelines and guidance documents are developed and updated by many international, regional and national organizations. Many of these documents are available online and can be accessed from the organizations' websites. If the NRA requires specific standards to be followed, then such standards should be made available to the BE. BEs should familiarize themselves with these documents, which are often updated regularly, to make sure that their own policies, processes and procedures are kept up to date.

5.4 Reference materials

Reference (measurement) standards are well-characterized preparations that are used as references against which biological products or laboratory assays are assessed. They play an essential role in the development of internationally agreed systems for measuring the biological and immunological activities of biological products. Their use supports the application of assays used in the standardization and control of biological therapeutics, which include blood-derived products. They are also used to ensure lot-to-lot consistency of production and to minimize systematic deviation of assays.

5.4.1 International standards

Many WHO Biological Reference Materials (BRMs) serve as reference sources of defined biological activity and provide a unique physical basis for the definition of WHO International Units (IU) that are used to quantify biological and/or immunological activity.

WHO International Standards (IS) are established by the Expert Committee on Biological Standardization and serve as the primary standard for the calibration of national and other secondary standards. A system of common standards is essential if the results of quality testing from different parts of the world are to be comparable. They are also essential when assessing and comparing the performance specifications of laboratory test kits during evaluation, for example, when selecting appropriate TTIA screening tests.

WHO BRMs are established and maintained through standard procedures *(52)* that include assigning a suitable unitage (often in IU) and assuring continuity and consistency in the biological activity of the unitage through consensus calibration of replacement candidates for existing BRMs. Some of the WHO BRMs are not established as IS but are provided as WHO Reference Reagents, which may act as interim standards. More information on WHO BRMs is available on the WHO website *(53)*.

5.4.2 Secondary standards

Secondary standards are reference materials that have been established by comparison against a primary standard. Primary standards (such as the WHO IS) are usually limited in supply and distributed only to national control laboratories and manufacturers of biological medicinal products. Regional and national authorities may therefore prepare and establish their own secondary reference standards, which are calibrated and traceable to the primary standard. Manufacturers of biological products or laboratory assays may also establish laboratory reference standards for routine use in assays, and these standards are calibrated by direct comparison with the IS or with a regional or national reference standard. Further information on regional and national regional standards is provided in various WHO publications *(52,54,55)*.

5.4.3 Certified reference materials

Certified Reference Materials (CRM) are batches of chemical materials used to check the accuracy of the equipment used for analysis in testing laboratories. These materials provide "measurement benchmarks" for materials testing and chemical analysis. They are used to ensure reliability and comparability of measurements as well as in establishing traceability. The use of CRM is a basic requirement in QA as noted in ISO 17025 *(56)* and described in the *ISO Guide 33 Use of certified reference materials (57)*. The *Code d'Indexation des Matériaux de Référence* is an international database for CRM that provides an Internet-based information service to assist testing and analytical laboratories in finding the CRM they need *(58)*. The ISO Committee on Reference Materials encourages its use as the basis for an international information system on CRM.

5.4.4 QC materials

The term QC as used in the *WHO Laboratory quality management system handbook (59)* means the use of control materials to monitor the accuracy and precision of all the procedures associated with the examination (analytical) phase of testing. QC materials are substances that contain an established amount of the substance being tested. The inclusion of QC materials to demonstrate that the laboratory tests are performing as specified is stipulated in most national and international standards including the *WHO guidelines on good manufacturing practices for blood establishments (4)*.

The BE should establish a QC programme for all tests conducted in its laboratories to ensure the accuracy of the test results, including TTIA screening and immunohaematological tests. The type of QC materials and how the results are used, analysed and monitored depend on whether the laboratory method is quantitative, semi-quantitative or qualitative. In quantitative tests, QC materials should be tested at the same time as the donor or patient samples. For qualitative tests, QC materials should be included with every screening run or batch of samples so that trends can be identified and addressed.

QC materials may be purchased *(60,61)*, obtained from reference laboratories or prepared in-house. When considering whether to purchase, source or prepare their own QC materials, BEs should consider the essential requirements for such materials (Box 5.1).

Box 5.1 Important requirements for QC materials *(59)*

- They must be appropriate for the test and the amount of substance being measured must be present in a measurable form.
- The amount of analyte in the control should be close to the critical decision points of the test. In quantitative tests, both high and low values should be checked. In qualitative and semi-quantitative tests, the positive control should include a control that is close to the cut-off value of the test.
- The QC materials should be validated and traceable to an international standard.
- The QC materials should have the same matrix as donor and patient samples, i.e. they should be based on serum, plasma or whole blood depending on the test requirements.
- The materials should be available in sufficient quantities so that they can be used for monitoring and should be stable during storage. To avoid analyte degradation, frozen control materials should generally not be thawed and refrozen.
- Most QC materials are prepared from patient samples. Unless they have undergone inactivation, they must be handled and stored as biohazardous material.
- Calibrators provided in test kits are used to set or calibrate the test kit or system and should not be used as QC materials.

Implementation of QC should include the establishment of criteria for acceptance of results based on specified limits of acceptability. The method of analysis and monitoring of QC results will depend on whether the results are quantitative, semi-quantitative or qualitative. The WHO handbook *(59)* provides useful information and guidance on the use of different statistical methods to monitor laboratory test performance. Whichever method is used, it is important that the BE has guidelines and processes in place to ensure that QC results are properly reviewed before any test results are issued. A system should be in place to recognize, identify and solve QC problems in a timely manner. QC data should be regularly reviewed by laboratory supervisors, the QM and management.

CHAPTER 6

Documentation

Documentation is an essential part of the quality system that ensures work is performed in a standardized and consistent manner as well as providing traceability of all steps undertaken in a process (Box 6.1). The BE will need to define the structure and operation of its documentation system and establish a master list of documents. BEs may choose to follow a documentation pyramid structure with managerial documents appearing at the top and operational documents, including forms and records, at the bottom.

The following are all important parts of the documentation system, but the list is not all-inclusive: mission and vision statements; written policies; a quality manual; process descriptions; SOPs or work instructions; forms; records; organizational charts; job descriptions; and labels.

Box 6.1 Benefits of a good documentation system

- provision of a framework for effective communication throughout a BE;
- prevention of errors that may arise from misunderstanding a verbal instruction;
- performance of processes and procedures in a standardized and uniform manner;
- orientation and continuing training of staff;
- full traceability of details;
- facilitation of audits and resolution of defects, complaints and adverse reactions; and
- means to communicate information to management about the activities of the BE.

6.1 Policies

According to *ISO 9000-2015 (62)*, a policy is "a documented statement of overall intentions and direction defined by those in the organization and endorsed by management". Policies describe the manner of operation of the BE and the direction for achieving its goals.

The quality policy is a brief statement that aligns with the BE's purpose, mission and strategic direction. It contains the quality objectives and includes a commitment to meet applicable requirements as well as to continually improve.

6.2 Quality manual

The quality manual is an important document at the policy level and can be defined as a roadmap explaining how the elements of the quality system apply and relate to each other. It should be written with the input of policy-makers, experts in the field of transfusion medicine and quality department staff. It should be revised, kept up to date and improved continuously. All staff should be trained in the appropriate use of the quality manual.

The quality manual describes the BE's quality system, including the organization structure, the quality policy and quality objectives, standards, processes and procedures. It briefly outlines the policy for every element in the quality system. Repetition of information contained in the SOPs should be avoided to prevent having to update the manual whenever individual SOPs are updated. The manual should be as concise as possible and tailored to fit the particular BE.

6.3 SOPs

SOPs are an essential part of the quality system *(63)*. They describe and explain activities and procedures step by step. SOPs should be detailed but concise, clear and easily understood, especially by new staff who are in training. SOPs should have an easy-to-read format and use of a template for the preparation of SOPs is strongly recommended (Box 6.2).

Flowcharts may be used in support of SOPs. They show the major steps in a procedure in a simplified manner, and include input, interrelated or interacting activities, and output. The activities explain the transformation of input to output and may be a useful starting point in planning and writing SOPs.

Any task aids used should be controlled in the same way as SOPs and linked to their overarching SOPs by including a reference to the SOP document codes. A task aid could be an excerpt from an approved SOP, an instruction card, a reminder sheet, a checklist or other similar tools.

Box 6.2 SOPs should contain the following information

- short title;
- aim and/or purpose of the activity or procedure;
- scope or departments involved;
- references;
- terms and definitions or glossary;
- materials required;
- step-by-step instructions for carrying out the procedure, which are written in sufficient detail and are easy to understand;
- staff responsible for each step in the procedure; and
- a unique document index number or code, version, date of approval or revision, effective date and duration of validity.

6.3.1 SOP preparation

Fig. 6.1 shows the process of SOP preparation, approval and release.

Fig. 6.1 Flowchart showing standard operating procedure (SOP) preparation, approval and release

Staff often have an aversion to paperwork as they consider it to detract from performance of their main tasks. Box 6.3 provides helpful tips on SOP preparation. To make documentation more appealing, it needs to look attractive and be simple to work with. SOPs should be prepared by knowledgeable staff with expertise in the procedure concerned, and who are involved in the BE's internal structure. SOPs should also be periodically reviewed and updated if there has been any change in the materials, methods or equipment used. Involving staff at various levels within the organization in the writing and reviewing of SOPs will enhance their commitment to them *(14)*.

Box 6.3 Helpful tips for SOP preparation

- Select teams of staff to write SOPs (and other documents) including representatives from each work area.
- Choose staff with a natural aptitude and a good command of the written language.
- Ask staff who are meticulous to review and critique the first drafts, usually including quality staff.
- Train the document writers and reviewers.
- Emphasize the importance of planning when writing SOPs by mapping out the process first.
- Set aside time for document writing sessions.
- Provide a location where the SOP writers will be undisturbed and supply refreshments.
- Start the session with a motivational video or talk, or show examples of excellent documents.
- Set target dates for completion of first drafts and challenge staff to meet them.
- Reward positive efforts and good outcomes.
- After producing the first drafts, encourage staff to streamline them further.

6.3.2 SOP contents

BEs should aim to keep SOPs short, sharp, specific and up to date (Box 6.4). Shorter documents mean faster processing time, less use of paper and a greener environment. Where possible, BEs should consider moving their documentation system to digital document control and only printing documents when necessary.

Box 6.4 Practical tips on SOP contents

- Set narrow margins and compact spacing to fit more words on a page.
- Break text into blocks under concise subheadings.
- Start each instruction with a verb, e.g. check, test, record, mix, add.
- Avoid jargon.
- Number each point with a simple numbering scheme – try not extend beyond three decimal points.
- Use bullets to list points.
- Number each page with the page number and total number of pages, e.g. page X of Y.
- Avoid repetition of the same text in multiple documents as this makes it difficult to track updates.
- Limit the number of signatories to three or four: expert/author, department head, quality department representative, (sometimes) chief executive/director.

6.3.3 SOP implementation

SOPs and related documents must be properly implemented and followed to achieve the desired impact. Draft documents need to flow through the system as quickly as possible, and the time allocated for each signatory to review and sign the document should be limited.

An important step before any SOP is released for use is validation of the process. That is, carrying out the procedure exactly as documented and checking that in doing so, the correct result is obtained. The validation of computer systems is especially important. Further information on validation of SOPs can be found in the International Society of Blood Transfusion (ISBT) Science Series book, *Introduction to blood transfusion: from donor to patient (64)*.

Staff need to be educated and trained in the implementation of new SOPs or following significant changes to existing SOPs. Superseded SOPs need to be retrieved and destroyed by the quality department.

The BE should institute a signing system for users to record that all relevant new or updated documents have been read and understood and that training has been received, so that only those with demonstrated competency carry out procedures without supervision.

6.4 Labels

Labels identify and describe what has been produced. Labels should be created, validated, maintained and managed within the document control system and should meet regulatory requirements. A master file is recommended for current labels. Reprinting new labels and discarding unused or surplus labels are critical activities and should also be controlled. Preprinted labels should always be properly accounted for and securely stored. Unused labels should be destroyed.

Validation of labels is an important aspect of document management. Upon receipt, preprinted labels should be checked against a proof to ensure that they contain the correct information. The durability and suitability of the label material and adhesive should be evaluated to ensure that they are safe and appropriate for use, particularly labels used for blood bags. Adhesive used on the label should be suitable for the storage temperature of the component (for example, labels attached to frozen plasma products). Use of unsuitable adhesive may result in the label becoming detached from the bag.

6.5 Forms and records

Forms are used to collect data of various kinds and may be paper-based or electronic. Once a blank form is filled with data, it becomes a record. Records provide objective evidence of activities performed or results achieved.

Each activity that may affect the quality of blood and blood components should be documented and recorded at the time it takes place. In the laboratory, recording test results accurately and immediately is important to avoid errors and ensure that results are linked to the correct sample. All records should indicate the person performing the action, the date (and sometimes the time) of the action and the equipment used in the action, where applicable.

Records should be legible, accurate, reliable and a true representation of the results and data entries. There must be mechanisms to identify corrections, deletions, omissions or modifications and the person responsible for such changes. Measures need to be in place for ensuring confidentiality of sensitive information relating to donors, patients and staff.

Records should be retained for a period that complies with national regulatory requirements. Storage conditions should ensure security and protection from damage.

It is essential that all the steps in the blood transfusion chain can be traced, both prospectively and retrospectively. Where the BE is not directly responsible for all the steps in the blood transfusion chain, it should be responsible for traceability of those steps that it is involved in.

6.5.1 Implementing a record system

The BE should aim to limit the number of records and the length of each document to the essential minimum. Only the information that is needed in terms of organizational or regulatory requirements should be recorded. The BE should begin by making a list of all essential records in each work area and then implement them incrementally, as it is impossible to create all forms in one attempt.

Forms should be designed and developed in the specific work area where they are to be used. Since forms are a dynamic aspect of the documentation system, staff should be able to update and amend them if required. However, the job title of the person authorized to do this should be clearly defined and records should be maintained of the changes. Forms should also be version controlled so that only the current version is available for use. Periodic audits should be conducted to check that the forms in use match the latest recorded version. The quality department should have an overview of all forms to maintain consistency.

6.5.2 Designing forms

Well-designed forms can improve compliance with procedures and prevent mistakes. Certain staff may exhibit a flair for document design and their talent should be utilized. Forms should be designed to look appealing, with creative use of colour, different typefaces, blocks and tables (Box 6.5). The BE should encourage a sense of pride among staff in maintaining complete and accurate records.

Box 6.5 Helpful tips when designing forms

- Standardize the format for forms across all departments; consider using a style guide or templates.
- Place recurring features such as headings, logos and page numbers in the same place on all forms.
- Keep headings short and succinct.
- Visually group related information.
- Follow logical sequencing and numbering.
- Allow enough space for data entry; match fields to the type and size of input required.
- Use clear, easy-to-read fonts and avoid fonts that are too small or cramped.
- Design paper-based and electronic forms to look similar.
- Number forms in a consistent manner throughout the BE.
- Avoid duplication of information recorded in more than one place or work area.
- Review and improve form design regularly.

6.5.3 Linking forms and SOPs

It is important to link relevant forms to specific procedures. This can be achieved by referencing or listing the forms by number and title in the SOP and adding the title and unique document number of the SOP to the form. Whenever an SOP is updated, the associated forms should also be reviewed. It is not recommended to include forms as an appendix to an SOP as a small change to a form will involve updating the whole SOP.

6.6 Document control systems for paper-based and electronic documents

The correct handling of large amounts of documentation is challenging for any BE, but it is critical that all documents are complete, accurate, up to date and include relevant data. Document control systems are essential to allow BEs to ensure that only approved, current documentation is used throughout the establishment. This is important in preventing inadvertent use of documents describing obsolete instructions, processes or SOPs.

A document control system should be designed to handle the process of organizing, routing, tracking, authorizing and distributing all documentation relating to all activities of a BE. When planning the document control system, the BE should ensure that a management policy is applied throughout the life cycle of any document – whether it be a hard copy or accessible via an electronic document management system. Each document should go through a series of steps that include creation, storage, sharing and distribution, archiving and ultimately destruction. Each step should meet specific requirements which ensure that documents are managed in accordance with established standards.

6.6.1 Creating a document control system

The process of creating a document control system is shown in Table 6.1.

Table 6.1 Creating a document control system

Step	Activity
1. Document and workflow identification	• Start by identifying and classifying all documents to be managed by the system. • Pay particular attention to records containing sensitive data (e.g. health status of blood donors) for which adequate control is an essential component of the protection of personal data.
2. Establishment of ownership and quality standard	• Determine what standards need to be met by each document. Relevant regulatory requirements should be assessed. • Clearly define who is responsible for each document in terms of creation, approval and release.
3. Document numbering	• Identify each document in accordance with a unique numbering system that names and classifies documents according to their intended purpose (e.g. INST for instruction, PROC for procedures, SOP for standard operating procedures). • This will be of key importance for retrieving documents using the search functions in an electronic filing system during operational activities or in case of audit or inspection.
4. Document revision procedures	• Establish a procedure for the periodic and timely revision of documents. This should describe who is allowed to make revisions. • Put in place traceability measures to record the document name, responsible staff members, review dates and descriptions of the changes.
5. Document access management	• Ensure that the document control system incorporates security measures, especially when it comes to sensitive or personal information. • Implement access control measures so only authorized people can access documentation to prevent unauthorized access, use or modification of documents and records. • Ensure that any access to data stored in computer systems is granted by a single identification name and password. Implement secure back-up arrangements in all electronic data storage systems to protect against loss of or damage to data.
6. Archiving procedures	• Establish version control to ensure that only current versions of the documents are accessible and in use. • Implement processes to ensure that all superseded or obsolete files are either archived or, in some cases, destroyed on schedule. One copy of the document should be archived, and the other copies collected and destroyed. Destruction of the copies should be documented. The document numbering and classification method should include a means to label documents as archived or obsolete.

The document control system should extend to any documents and records stored off-site. If storage of these documents or records has been outsourced, clear arrangements must be made with the storage provider to ensure that the same processes and procedures for managing them as on the BE premises are followed and periodic audits conducted to check compliance. The period of retention for all documents and records stored within the BE or off-site should be clearly stipulated, taking national regulatory requirements into account.

Electronic document control systems can greatly improve document control and should be considered where they are available and accessible. For example, most of these systems can dynamically apply limitation statements on printed versions or even limit printing by unauthorized users.

CHAPTER 7

Training and staff development

Training is the process of providing and improving staff knowledge, skills and performance to meet the requirements of the job. Training of staff should be part of career development and linked to clearly defined career paths accompanied by appropriate supervisory processes, and objective performance-based appraisal systems. Sound human resources and workplace practices affect performance quality and should be established within the quality system.

Establishing a specific training department within the BE helps to provide the basis for a standardized and consistent approach. It also provides the opportunity to utilize training professionals, or for BE staff to be trained as trainers.

7.1 Identification of training needs

The need for training arises when there is an identified gap between the requirements of a specific job and the existing competency of the staff assigned to that job.

Training should be provided initially for new employees and followed by ongoing training and education thereafter. Before planning training or continuing education programmes, training needs should be identified. Performing a training needs analysis is the premise for determining training objectives and designing training plans, and also provides a basis for assessing effectiveness of the training. Table 7.1 describes the key steps in identifying training needs.

Table 7.1 Identification of training needs

Key step	Requirements
Identification of the knowledge and skills required for each job position	• Job descriptions (discussed in Chapter 4) should clearly and accurately state the level of education, knowledge, skills and experience required for each job, and the expected duties and responsibilities including quality-related ones.
Assessment of the competency of each staff member	• The competency of the staff should be assessed when they have completed initial training. For staff who are already working, competency may be determined through assessments of daily work performance. • Retraining should be provided when competency assessment reveals the need for improvement. Competency should be re-assessed after any job-specific training. • For blood establishments (BEs) establishing a quality system for the first time, all staff will require training in the basic principles of quality management, as well as training related to the specific operational requirements of the quality system that are relevant to their job.
Considering the need for organizational and staff development	• As the BE's business scope expands, the management should consider the need to provide training and career development for their staff as part of organizational development and to meet new job requirements in some positions. • For the staff, additional training will help them to achieve their own career goals. Providing appropriate training opportunities that will enable staff to improve and expand their competencies is a good motivation towards performance excellence and helps to foster a culture of learning and quality within the BE.
Aligning training needs with quality system goals	• Quality management is a process of continual improvement. As the quality system develops, training needs will also change and should be periodically reviewed. The review should be based on feedback from management and staff or observations from trainers or experts, either in person or through questionnaires. • The BE should institute a system for gathering and evaluating this information to determine whether the priorities and training provided will be adequate to achieve organizational quality goals.

7.2 Training policy and plan

7.2.1 Training policy

The BE should have a training policy that will support its mission and objectives. It should apply to all the programmes and activities of the BE, and should include the following *(3,4)*:

- All BE staff and other health-care professionals involved in any element of the blood transfusion chain should be comprehensively, appropriately and effectively trained.
- Staff should receive initial and continuous training that is appropriate to their specific tasks.
- Training should be carried out by qualified staff or trainers, where possible, and should follow pre-arranged written programmes. Ad hoc on-the-job training may be performed by experienced staff within the department concerned.
- Approved training programmes should be in place and should include the basic principles of transfusion medicine, quality system and GMP, relevant safety and hygiene practices, and relevant standard terms.
- Training should be documented, and training records should be retained.

The training policy should also set out specific procedures and performance standards such as:

- identification and documentation of specific staff training needs;
- documentation of financial resources available and utilized for staff training and development;
- development and approval of specific training programmes; and
- identification and documentation of training outcomes and any work-related improvements.

The training policy should be reviewed periodically at specified time intervals. A review may also be necessary if there are changes to legislative policy, the funding environment or operational activities that render the existing policy inappropriate or redundant.

7.2.2 Training plan

Planning of training activities is important to ensure that the training goals are met (Box 7.1). When planning a training activity, the BE should consider:

- the training needs that have been identified;
- budget and resources available for training;
- trainer(s) and training materials and tools to be used;
- design of training;
- methods for provision of training; and
- how training outcomes will be evaluated and monitored.

Training needs should be used to guide the objectives and content of the training programme, as well as the evaluation of training outcomes. Training programmes should be designed based on the budget and resources available. Training materials should be prepared before training commences. Any consumables and reagents that may be required during training should be of the same standard as those used for routine operations. It is important that all training plans and training activities are properly documented, and records retained (see also Box 7.1).

Box 7.1 Helpful tips for planning successful training events

- Keep training sessions short to minimize staff time away from routine tasks.
- Choose the location carefully, ensure adequate space, lighting and ventilation.
- Check that audiovisual equipment is available and operational, if it is required.
- Consider providing participants with notepads, writing materials and refreshments.
- Make training fun, link it to an interesting theme, add some humour.
- Include exercises and quizzes with small prizes or incentives.
- Involve participants by engaging them in role play.
- Make presentations and other training materials colourful and vibrant.
- Maintain an attendance record for each session.
- Arrange for staff to earn continuous professional development (CPD) points for attendance.
- Supply handouts, notes, posters or information on memory sticks for staff to take away.
- Solicit feedback from participants after each training session and use the information for future improvements.

7.3 Conducting training

To ensure that training is taken seriously by supervisors and staff and is effective, the BE should take steps to ensure the following:

- Formal orientation procedures are in place for all BE staff and volunteers.
- Basic training for all BE staff and volunteers is provided in accordance with the policy on blood transfusion services.
- Further training and development opportunities for individual staff are offered through formal supervision and the performance review system.
- Management training opportunities are available for senior BE staff.
- Attendance at conferences and other training and development programmes is encouraged wherever feasible.
- Training opportunities are properly researched, costed against budget parameters and promptly approved.
- Training and development records are maintained for all BE staff and volunteers, which include training goals, training undertaken and any post-training assessment reports (Box 7.2).
- Provision for staff training and development is made in the BE's annual budget.
- All staff are given an equal opportunity to participate in all training and development activities.

- Any staff grievance regarding training and development is promptly investigated, remedied and documented.
- Any training-related grievances that have been lodged are addressed in accordance with the training and development principles and procedures outlined in the BE's training policy.

Box 7.2 Training records *(4)*

- The records should identify the trainer, all the tasks the trainee is being trained in (including the relevant SOPs), and when the training was completed.
- Records should be signed by both the trainee and the trainer.
- Upon completing the training, the staff members should be assessed and found to be competent in the tasks in which they have been trained.
- Staff training profiles and records should be kept up to date, including those maintained on computer databases.

7.3.1 Training of all BE staff on quality

All BE staff should receive training in the general principles of quality, the quality system, documentation and the use of quality monitoring tools *(3)*. This applies to initial training and subsequent on-the-job training in all departments. The quality department should have oversight of the content of the training materials if the training is not performed by its staff. Training should be regularly carried out to update staff on new developments in the quality system.

7.3.2 Initial training

Initial training should be conducted:

- as part of orientation procedures for new staff;
- for existing staff taking on new functions; and
- for staff returning after an extended absence from the workplace.

The curriculum for initial training should cover all relevant tasks and procedures, including general topics such as QA, quality system requirements and computerized systems *(4)*. Initial training may also include practical training for staff in job positions associated with blood collection and laboratory work. The duration of initial training should be defined depending on the complexity of the job.

New recruits and staff taking on new functions should be fully competent in the tasks assigned to them before they are given authorization to work without supervision. Only qualified senior staff with the relevant experience should act as trainers, and a process to evaluate the trainers should also be established and documented.

7.3.3 Cross-training

Cross-training offers staff an opportunity to acquire skills outside their own areas of work or discipline. This also provides the BE with flexibility in staff scheduling or reassignment, which may prove critical in crisis situations or when staff are absent due to illness or are on leave. Training plans should be tailored to the experience and work assignments of the staff. It is often not necessary to repeat training on general topics if the staff have already been trained and are competent in these areas as part of their usual work.

7.3.4 Retraining

Retraining is required when competency assessment reveals the need to improve a staff member's knowledge and skills. If the results of the assessment clearly identify specific areas of weakness, retraining can be targeted at those areas. However, if lack of competency in many areas is indicated, the staff member should be required to undergo initial training again.

After the training, the trainees should be re-assessed and only those who have passed the competency assessment should be allowed to return to work. Regular retraining may also be required for tasks that are only occasionally performed, for example, a specialized component-processing activity or emergency procedures. This also applies to manual methods that might be occasionally used as back-up when automated tests are not possible.

7.3.5 Continuing education

Continuing education programmes (theoretical and/or practical training) help to keep the staff up to date with their knowledge and skills. Topics for continuing education may include any changes to SOPs and operational changes that affect the staff. Where relevant, continuing education programmes should include technical and scientific developments outside the BE. Both internal and external training courses may be useful *(4)*.

The minimum number of annual training hours for staff to attend continuing education programmes should be determined and provisions made to ensure that staff are able to attend these programmes. For some categories of professional or technical staff, local licensing regulations may stipulate a minimum number of hours for attendance of continuing education programmes.

7.3.6 Supportive supervision

Supportive supervision is a concept that may be considered within the national blood system. This involves regular visits by trained supervisors to blood centres and transfusion-performing health facilities, where they observe, monitor, mentor and provide feedback to staff on their performance and adherence to standards and protocols. Unlike external audits, supportive supervision focuses primarily on education and training activities conducted by external experts and trainers. These programmes are usually organized by the national blood programme manager or senior management of the BE. An effective supportive supervision programme emphasizes follow-up activities after the visits, establishes an effective mentoring mechanism and promotes teamwork.

By focusing on empowering local staff (including staff in remote and peripheral areas), supportive supervision provides a unique opportunity to ensure that quality standards are met in the BE and that there is continuous quality improvement. More detailed information on supportive supervision can be found in WHO documents and in manuals produced by national organizations *(65,66)*.

7.4 Education of other health-care professionals involved in blood transfusion

To ensure patient safety, the quality policies and measures need to be applied across the entire blood transfusion chain including the safe provision and transfusion of blood and blood components. Health-care professionals involved in blood transfusion should be trained in the basic principles of quality, including any BE quality system requirements related to their duties. Although it may not be directly responsible for health-care professionals outside its organization, the BE should consider facilitating or supporting training programmes to ensure that blood and blood products are handled and administered appropriately (Box 7.3). This may involve organizing training programmes, workshops, seminars or continuing education activities for other health-care staff, or senior BE staff may give lectures on topics related to transfusion medicine.

Box 7.3 Examples of topics for training programmes for other health-care professionals who are directly involved in in blood sampling, the decision to transfuse or blood administration

- knowledge of different blood groups and clinical practice implications;
- evidence-based indications for clinical use of blood and blood products;
- risks and benefits associated with blood transfusion;
- alternative blood replacement therapies and blood conservation strategies, and treatments available;
- critical checks that clinical staff must make before, during and after administering a blood transfusion;
- concepts of patient blood management;
- clinical transfusion practice and patient blood management (for clinicians who are responsible for the decision to transfuse and the clinical management of patients receiving blood transfusion).

If the BE undertakes the task of centralized testing or processing, and blood samples or blood donations are collected by other institutions, staff in the blood collecting institutions should be trained in the implementation of the appropriate SOPs for ensuring the quality of the blood and blood samples during collection, storage and transport. A system of quality monitoring should also be established with these institutions. If blood and blood samples are to be transported to the BE by an outsourced provider, the staff responsible for the sample delivery should be appropriately trained to ensure the quality of the blood and samples during transportation.

7.5 Evaluation of training and its impact

Post-training evaluation should be conducted after all training activities as part of continuous improvement and monitoring efforts. The purpose of the evaluation is to assess:

- whether the training meets the intended objectives;
- whether the training meets the expectations of the trainees;
- the competence of the trainer;
- the suitability of the training plan or programme and its delivery; and
- opportunities for improvement.

The design of the training plan or programme should include the methods for evaluating the effectiveness of the training. Typically, the objectives and expected outcomes of each training activity would be set based on the training needs of the trainees. The method used to assess whether the training has been effective in meeting these objectives will depend on the type of training that has been conducted. For example, if training primarily involves knowledge transfer, then the level of knowledge of the trainees can be assessed through written or oral quizzes and tests to see if it has improved. Training involving more practical hands-on skills may be evaluated through practical competency testing. Feedback obtained from the trainees, through questionnaires or verbally, will help in assessing the competence of the trainer as well as the suitability of the plan and the training programme.

The long-term impact of training programmes can be assessed by periodic reviews of staff competency assessments, including routine observation of staff performance. QIs can also be developed and monitored during quality or management reviews to evaluate and monitor the impact of training (Box 7.4).

Box 7.4 Helpful points to monitor the effective implementation of training programmes

- All new employees have been provided with a copy of the BE's policy on staff training and development and a staff copy of the policy is kept in each service area.
- New staff and volunteers have successfully completed the initial training after appointment.
- All BE staff have a documented training and development plan within 12 months of appointment.
- Identified training and development activities have been completed within an agreed time frame.
- Staff training and development records for all BE staff have been maintained and updated.
- An annual review of performance documentation has been undertaken, demonstrating routine periodic and timely feedback on performance and outcomes.
- All training and development activities have been recorded on individual staff records, which have been collated and reported to management on a regular basis.
- Any training-related grievances that have been lodged have been addressed.

7.6 Leveraging existing resources

BEs may not always have the resources available to conduct all their training themselves, due to limitations on finances, space or human resources. Sometimes the BE may also not have the expertise in-house to conduct full or comprehensive training in more specialized topics. In such situations, the BE should consider how best to utilize existing resources or leverage external resources.

7.6.1 Resources in the BE

Training activities can often be organized at low cost with the resources available in the BE. Examples include starting a journal club or discussion group, holding quality events or organizing educational sessions using available audiovisual resources. Staff could be encouraged to research topics and present their findings to colleagues or to participate in interactive self-study programmes such as e-learning freeware or printed courses. Senior BE staff could be encouraged to collect and maintain a set of teaching slides and to give periodic talks on their areas of expertise to junior staff.

7.6.2 Local resources

When organizing internal training activities, the BE should consider local expertise available in the health-care community. Local experts can be invited to give lectures, conduct continuing education activities and participate in information exchange discussions. Some of these resources include QA committee, experts in laws and regulations, clinicians, nurses, haematologists, immunologists, pathologists, infection control personnel, epidemiologists or surveillance officers, local manufacturers and external assessors.

7.6.3 External resources

External continuing education programmes can also be led by international topic experts, such as those associated with other blood services, proficiency testing services, manufacturers, scientific societies, WHO, nongovernmental organizations or regional networks. Many organizations also make training materials and programmes freely available on their websites or organize training seminars and workshops online.

CHAPTER 8

Assessment – management of process and improvement

One of the key elements of the quality system in a BE is continuous assessment of the effective implementation of the various strategies to ensure availability of safe blood for transfusion. This ongoing assessment includes strategies to manage and improve processes. The assessment involves:

- evaluation of available equipment and reagents to select the most suitable ones, including feasibility studies where appropriate;
- validation of all processes, procedures, equipment and reagents;
- implementing a system for change control;
- reporting and analysis of errors, accidents and incidents followed by effective corrective and preventive action; and
- regular review of all activities to assess the overall effectiveness of the quality system and ensure continuous improvement.

8.1 Validation of processes, methods, equipment and reagents

Validation refers to the process of ensuring that a particular process, item of equipment or reagent will perform as expected. It is that part of the quality system that assesses in advance the steps involved in managerial and operational processes and procedures to ensure consistency (reproducibility) and quality, effectiveness and reliability. Objective evidence that results in a "high degree of assurance" that the process is in fact meeting expectations is required.

A structured approach to performing validation includes the three steps described in Table 8.1.

Table 8.1 Steps in validation

Step	Activity
Planning the validation	• All validations should be planned. • The planning team should include experts in the area to be validated. • The validation plan should include timelines for implementation. • The key inputs for each process should be defined, qualified and fit for purpose.
Performing the validation	• All validations should be performed by staff familiar with the processes being validated.
Preparing the validation report	• A formal report should be prepared after the validation has been conducted. • The format for validation documentation should be standardized for all validation activities to ensure consistency.

An approved written plan or validation protocol is an essential part of putting a new process, item of equipment or reagent into routine use (Box 8.1).

Box 8.1 Helpful tips for validation documents

The validation plan should include the following:

- purpose and scope
- definitions
- background and references
- validation approach
- implementation strategy
- training requirements
- document change control sheet – needed for changes in a reagent, procedure or method, equipment or the environment.

The validation summary report should include:

- validation plan details
- validation results
- unexpected results or problems
- recommendations and further actions
- details of continuing validation or monitoring activities
- references
- training requirements related to responsibilities.

Note: In addition to the validation report, all raw data generated during validation activities are to be kept on file as evidence.

8.1.1 Process and method validation

A test method must be shown to be fit for purpose so that end-users have confidence in the results it produces. Method validation provides objective evidence that the method is fit for purpose, which means that the requirements for its specific intended use are fulfilled *(67)* (Box 8.2). Method validation is an essential requirement for ISO/IEC 17025 *(56)*

Box 8.2 Examples of processes and procedures that require validation

- test methods;
- blood component preparation and storage;
- blood component labelling and tracking;
- sample handling, especially if robotic dispensing systems are used;
- controlled temperature storage of critical reagents and controls; and
- reporting of results.

There are two steps in method validation:

- Step one is to specify what the BE intends to identify or measure, both qualitatively and the quantity (if applicable).
- Step two is to identify the performance parameters for validation.

It is important to ensure that the sample sizes used during method validation are large enough to produce statistically valid results.

8.1.2 Equipment validation

Equipment validation should cover all the components involved, namely, hardware, software, consumables and reagents. The validation process includes three specific "qualification" steps:

- **Installation qualification (IQ):** refers to compliance with environmental requirements, safety features including electrical checks, and calibration.
- **Operational qualification (OQ):** includes equipment configuration and settings, self-checks, verification of sample volumes, alarms, safety features, connectivity with other IT systems, data sharing, barcode reading/ sample identification and reagent performance.
- **Performance qualification (PQ):** running the equipment in parallel with the current system. This should include testing of all methods, interpretation, back-ups, interfaces and data archiving to ensure accuracy and consistency.

8.1.3 Reagent validation

Reagent validation is important to ensure that the sensitivity and specificity of the reagent is appropriate and that it performs consistently. This minimizes the chances of false-positive or false-negative results, which affect the safety of the blood products and test results provided by the BE.

Reagents used by the BE should include the appropriate documentation to ensure that they meet quality requirements. This includes appropriate certificates of analysis from authorized manufacturers, expected performance data and specified environmental requirements for storage, for example, temperature and humidity. Package inserts should be provided with each reagent providing the relevant information on its expected performance and instructions for use. Where possible, reagents should be CE marked or equivalent.

Reagent validation involves testing them using appropriate samples to ensure that each reagent performs as expected in the BE environment (Box 8.3).

Box 8.3 Validation testing for reagents

- Samples used should be as close as possible to those used during routine operations in the BE.
- Quality control materials or sensitivity/specificity panels may be included if indicated.
- The number of samples to be tested should be predetermined to ensure that the sample size is sufficiently large to achieve statistical significance.
- If the analyte is infrequently detected, the BE will need to balance the need for a sufficient sample size with the associated costs and logistic constraints involved.
- If the test is already being implemented, the same samples as are being tested by the current system may be used so that the performances can be compared.
- If the test is already in use, BEs will have to decide whether to test the new reagent in parallel with the current system or at different times. The implications of the choice made will need to be carefully considered, particularly in the case of TTIA screening reagents.
- BEs may also consider batch validation for critical reagents.

8.2 Change control

A change is a permanent planned alteration to any approved process or procedure that may affect the safety, quality and efficacy of the process or procedure involved. Change control is a formal system to evaluate, document, approve and implement changes that might affect the validated status of facilities, equipment and processes *(69)*. It minimizes risk to an organization.

There should be a policy in place defining the process to be followed when making a change. Proposed changes should be reviewed and approved prior to implementation. A formal assessment of the impact of the change on validated processes, including the risk assessment, should be made. All the changes should be recorded. The documentation should include the type of change, processes to be followed, personnel authorized to make the change and time frame for the change.

A post-implementation evaluation should also be carried out to determine whether the change has been effective and successfully implemented.

8.3 Error management, corrective and preventive action

Incidents and errors, which are sometimes also called non-conformances, non-conformities or non-compliances, occur in every organization. They can be reduced, eliminated and prevented only when they are reported as and when they occur. Note that not all incidents are errors.

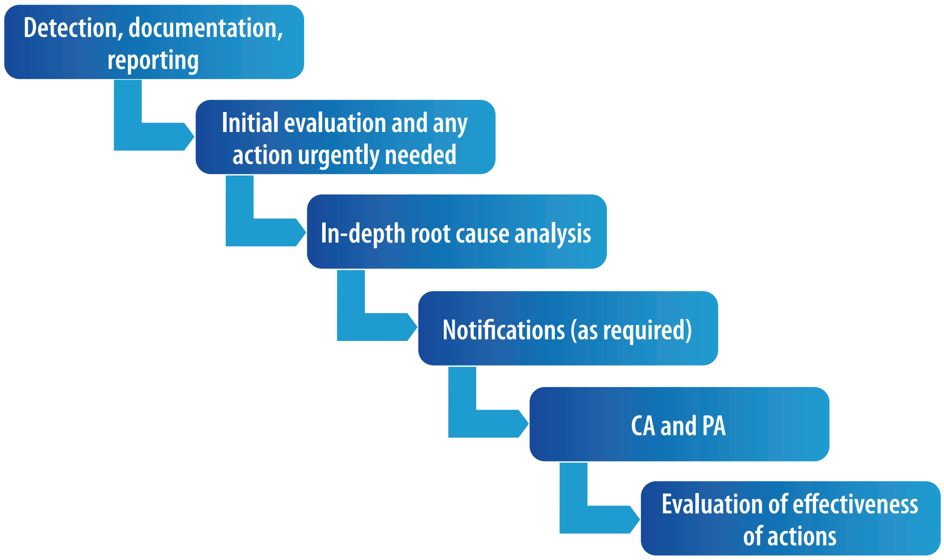
Near-misses, on the other hand, occur when actions that could have resulted in harm, loss or damage were fortunately avoided. In the BE context, near-misses that could have resulted in harmful consequences to the patient are those that were identified prior to transfusion. Near-misses reflect the true incidence of errors and should be documented and reviewed as part of the quality improvement process. Errors can occur at any point in the transfusion chain and BEs should ensure that a system is in place to recognize errors and incidents.

8.3.1 Error and incident management

Error and incident management in the BE is core to continuous process improvement. It requires a well-designed process by which errors and incidents can be identified without apportionment of blame and handled effectively. Error reporting should not be the final goal, but only a means of learning to help improve processes and procedures.

Error and incident management should incorporate both corrective action (CA) and preventive action (PA). CA is the action to eliminate the cause of a detected non-conformity or other undesirable situation to prevent a recurrence. It is reactive and resolves the immediate problem, whereas PA is proactive action to eliminate the cause of a potential non-conformity or undesirable situation through planned activities (Fig. 8.1).

Fig. 8.1 Process for the management of errors and incidents



CA, corrective action; PA, preventive action.

A risk-based approach including risk assessment should be taken. The systematic acknowledgement, recognition and analysis of errors is an integral part of error management.

The use of root cause analysis (RCA) to investigate incidents and errors provides a structured approach to their investigation. This will ensure that opportunities to improve systems and processes are identified and adopted. When properly undertaken, RCA will encourage staff to report incidents and errors, whereas a system that focuses on blame is likely to result in non-reporting. This does not mean that individuals should not be held accountable for their actions when appropriate. Repeated errors by the same individual need serious discussion and appropriate measures should be taken (Table 8.2). This approach also acknowledges that most people willingly blame themselves for their actions and need to be supported.

Table 8.2 Root cause analysis

Root cause	What occurs	How to manage	Approach to staff
Human error	Inadvertent action: slip, lapse, mistake	Changes to • environment • processes • procedures • design • training	Support
At-risk behaviour	A choice: risk not recognized or believed to be justified	• Remove incentives for at-risk behaviours • Create incentives for healthy behaviours • Increase situational awareness	Coach
Reckless behaviour	Conscious disregard of unreasonable risk	• remedial action • punitive action	Sanction

The way errors and incidents are investigated and managed is often a reflection of the level of commitment that managers have to quality in the BE. Incidents should be investigated in an honest and open way and seen as opportunities for system improvement rather than the fault of the people involved. This is part of the principle of system accountability. Norman, in his book *The design of everyday things (70)*, writes that "People make errors, which lead to accidents. Accidents lead to deaths. The standard solution is to blame the people involved. If we find out who made the errors and punish them, we solve the problem, right? Wrong. The problem is seldom the fault of an individual; it is the fault of the system. Change the people without changing the system and the problems will continue".

8.3.2 Error prevention

Effective PA, as part of error management, will help to reduce future risks. BEs should establish policies, procedures and documentation for error prevention. Errors commonly occur due to failure to follow procedures, inadequate training and incomplete documentation. Availability of clear SOPs, work instructions and training of technical staff, implementation of correct identification protocols for donors and blood units, use of bar code technology and automation are some of the measures that can help to reduce errors.

The goal of error prevention is to identify potential risks, problems or incidents before they happen. Error prevention investigations will help to identify recurring incidents, gaps and weaknesses in existing processes and procedures. Competency gaps in the staff that can be remedied with training or retraining can also be identified. Table 8.3 describes the steps to be taken in the error prevention process. This process should be supervised by the quality team and reviewed during the management review.

Table 8.3 Steps in error prevention

Step	Activity
1	Identify the problem. • Events may be discovered during record review, staff observation, while performing a task, during internal or external audits, or through notification of a deviation by blood users.
2	Initiate immediate remedial action(s) to rectify the problem and prevent any dangerous consequences of the error.
3	Conduct a thorough investigation. • Understand how or why the problem occurred and the circumstances at the time of occurrence. • Include an evaluation of the environment, such as work conditions, workflow, facility problems, equipment and reagents, relevant standard operating procedures (SOPs), work instructions, staff training and performance in accordance with defined SOPs, and any recent change in processes.
4	Do a root cause analysis. • Ask the questions "what, how and why?" in a structured way. • Look for the actual cause of the problem rather than simply dealing with its symptoms. • This will help to determine the appropriate corrective and/or preventive actions and prevent problems from recurring.
5	Evaluate the effectiveness of corrective and/or preventive actions by regular follow-up. • This may include tracking and analysis of performance trends.

8.4 Management review

Periodic management reviews should be undertaken to monitor the effectiveness and the operation of the quality system, and to identify opportunities for continual improvement of BE systems and processes. Management reviews should always involve the senior management and the QM.

The requirement for management review is clearly stated in the ISO 9001 standard *(24)*. The standard stipulates that the management review process requires senior management to periodically review the quality system to ensure its continuing suitability, adequacy and effectiveness, as well as addressing the possible need for changes to quality policy, objectives, targets and other elements of the quality system. The review includes the assessment of opportunities for improvement and the need for any changes to the quality system. Where indicated, this includes updating the quality policy and quality objectives.

A management review has significant benefits beyond meeting the requirements of regulatory authorities. It enables management to review data on the inputs and outputs of all processes and procedures and make informed improvements that align with quality objectives. Table 8.4 lists the inputs and outputs of a management review.

Table 8.4 Inputs and outputs of a management review

Area	Items
Inputs	• Follow-up actions from previous management reviews • Assessment of risk management actions • Audit and regulatory inspection reports • Complaints, non-conformances, returns and recalls • Identification, review and control of incidents • Status and effectiveness of the corrective and preventive action system • User feedback • Staff suggestions • Quality indicators • Performance in external quality assessment • Performance of suppliers • New or updated regulatory requirements • Quality policy and quality objectives Other areas for periodic and timely review may include (but are not limited to): quality standards and metrics; specimen collection; staff training and competency; maintenance and calibration of equipment; internal quality control procedures; performance in proficiency testing; and the status of the document control system.
Outputs	• Recommendations for improvement of the system, processes, procedures, products or services • Identification of the need for resources

Records of the management review should be kept. The results should be communicated to all staff, and any follow-up actions implemented within an agreed time period.

8.5 Quality review

The quality review is an important quality management process required in most international guidelines on quality systems. It provides an opportunity to assess how well the organization is doing and to identify areas for improvement. Senior managers should be seen to be actively involved in, and ideally should lead, these processes. Periodic quality reviews should be performed to maintain an effective quality system and continuously improve its performance.

Quality review also provides senior management with the opportunities to identify areas for improvement in their management systems. It is important to understand the underlying principles of quality review requirements so that BEs not only meet the requirements of the GMP, ISO or other quality standards, but also use the periodic review process as an opportunity for collective reflection on continuous improvement.

8.5.1 Product quality review

Product quality reviews (PQRs) are mandatory to fulfil a compliance requirement, while also giving real insight into the performance of critical aspects of the quality system and providing a valuable platform for continuous improvement (Box 8.4). The requirement to perform regular PQR is present in most codes of GMP including the *WHO guidelines on good manufacturing practices for blood establishments (4)*. PQRs are designed to verify the consistency of existing processes and procedures, and the effectiveness of control strategies that are in place. The intent of these reviews is to perform an internal appraisal of all facets of the collection, testing, preparation, storage and use of the blood products, to highlight any significant trends in quality system data and performance, and to identify the need and opportunities for product and process improvements. Such a review is conducted at least annually and should be documented.

Box 8.4 Examples of aspects that may be considered in a PQR

- starting materials;
- critical in-process controls;
- results of quality control and quality monitoring;
- all changes;
- validation status of equipment;
- technical agreements and contracts;
- all significant incidents, errors and the corrective actions implemented;
- findings of internal and external audits and inspections, and the corrective actions implemented;
- complaints and recalls;
- donor acceptance criteria;
- donor deferrals; and
- look-back cases.

High-quality data will need to be collected and analysed from many processes (for example, collection, testing, storage, distribution, non-conformances, deviations, CAs, complaints and recalls) and sources (for example, blood donors, material suppliers, BE staff and BE customers). Conducting a PQR may be new for a BE that is working to meet GMP requirements, and it should take the opportunity to develop and strengthen its data management system and data analytics capabilities. Some BEs may already have started analysing and acting on strategic-level data on a regular basis, and it is important to link and integrate the quality review activities with activities that are already taking place.

Helpful tips on PQR can be gleaned from the *PIC/S Good practice guidelines for blood establishments and hospital blood banks (71)* and the *Council of Europe EDQM good practice guidelines* contained in the *Guide to the preparation, use and quality assurance of blood components (41)*.

CHAPTER 9

Assessment – monitoring of performance

In addition to managing and improving processes and procedures as described in Chapter 8, the BE should put in place programmes to continuously monitor and assess its performance. This ongoing assessment includes:

- a programme of regular internal and external audits of the quality system;
- active participation in appropriate EQAS to improve laboratory performance;
- selection and use of QIs for measuring and monitoring performance;
- continuous collection and analysis of data generated from key activities and their use in quality improvement; and
- establishment of haemovigilance through a system of monitoring, reporting and investigation of adverse incidents related to all blood transfusion activities.

9.1 Internal and external audits

According to ISO 9000-2015, an audit is a systematic, independent and documented process for obtaining objective evidence and evaluating it objectively to determine the extent to which the audit criteria are fulfilled *(4,14,62)*. Box 9.1 summarizes the benefits of audits.

Internal audits (also called self-assessment or first-party audits) are performed by trained staff from different departments within the same BE, who are not responsible for or involved in the processes or procedures being audited. Internal audits are useful for flagging issues or identifying areas in the BE that require more focused quality improvement efforts. Audits should be performed periodically at a frequency determined through a risk-based approach.

External audits are performed by staff from organizations outside the BE. Second-party audits are carried out by customers or suppliers, while third-party audits are carried out by organizations that provide certification or registration of conformity, or governmental agencies *(14,62,72)*.

Box 9.1 Benefits of audits *(3,14,59)*

- Facilitate and inform continuous improvement in the BE.
- Provide independent evaluation of systems and processes.
- Ensure adequate documentation.
- Evaluate understanding and implementation of BE policies, processes and procedures among staff.
- Assess staff competence and training.
- Support identification and discussion of problems.
- Enhance communication between management, staff and auditors.
- Improve staff motivation and help to create a quality culture.
- Help to generate mutual confidence and trust among BE staff, stakeholders and customers.

9.1.1 Establishing an audit programme

An effective audit should follow the steps shown in Fig. 9.1. All audits should be carried out by trained and competent certified staff in accordance with relevant international or national standards and guidelines, regulations, legal requirements, SOPs and approved procedures. Audit findings should be clearly documented in an audit report. Based on the results of the audit, appropriate CA and PA should be developed, agreed upon and implemented within specified time frames and with clearly defined outcomes *(12,59,72)*.

Fig. 9.1 Steps in an audit

Audit plans should be prepared in advance and cover all activities in the areas being audited. In most situations, the audit should be scheduled and the auditee informed in advance. On occasion, an external audit may be carried out without prior notification, so it is important that BEs always maintain consistent quality system performance, rather than just before an audit *(14,59)*.

Audit procedures should be performed according to the audit plan, and the auditor should be familiar with relevant standards as well as all processes, procedures and functions of the area or organization being audited. The availability of a detailed checklist can help auditors to conduct the audit more effectively *(12,14,72)*.

Audit reports should be prepared after the audit and according to the audit schedule. Deficiencies identified during the audit should all be linked to a clause in a standard or SOP. These deficiencies are usually classified as critical, major or minor and require CAs to be taken by the BE. The urgency of the CAs will depend on the risk involved. Areas of weakness marked as suggestions or observations do not usually require formal CAs to be reported by the BE, but provide useful feedback for quality improvement. Audit reports should be shared and discussed with the auditee, as well as with the quality department and management. Audit reports are confidential and their dissemination should be controlled *(4,20,72)*.

Audit follow-up should involve a review of the audit report by the BE staff involved in the processes being audited. Auditees should always be given the opportunity to respond to the report and to notify the auditor of any CAs that will be taken (if any) and their timelines. At the end of the agreed timelines, a review should be conducted to determine the effectiveness and impact of CAs taken. In the case of internal audits, this review may be conducted within the organization by the quality department and management, or, in the case of external audits, with the auditors. Table 9.1 details the important components of each step of the audit process.

Table 9.1 Components of the audit process

Step	Components
Audit plans *(59,72,73)*	• Objective and scope • Date and time • Inspection team and roles of members of team • Standard to be used for the audit • Documents to be requested before and during the audit • Requirement that auditee should be available during audit • Timetable for opening and closing meetings • Time required for preparing and communicating the report
Audit procedure *(72,73)*	• Opening meeting – to introduce the audit team and the purpose and scope of the audit, review changes, and evaluate the corrective and preventive actions taken to address the deficiencies identified by the previous audit • Tour of the facility – according to the flow from starting material to final products in the BE • Review of: management system, documentation, quality assurance and quality control results • Review of validation, verification, calibration and maintenance procedures and records • Sharing of audit findings – including positive and negative observations and areas for improvement • Closing meeting – to review the audit findings, discuss any deficiencies and plan immediate actions to be taken in response to critical deficiencies
Audit reports *(14,72,73)*	• Name of auditor, auditees and observers • Objective and scope of the audit • Date and location • Details of audit plan • Identification of audit standard/criteria • Classification and description of deficiencies, and link to corresponding requirement • Evaluation of quality system • Proposal for corrective and preventive actions and the relevant timeline • Follow-up time frame for proposed corrective actions • Date of submission for corrective action • Conclusion
Audit follow-up *(4,12,14,59)*	• Review of deficiencies and suggestions • Root cause analysis • Plan of corrective and preventive actions to be taken, which should include a timeline and the department(s) and staff responsible for the action • Monitoring progress of corrective actions • Record of all actions, results, responsible person, starting and ending date of corrective actions, confirmation and approval of corrective actions • Report for quality assurance department, management and auditor

9.2 External quality assessment

External quality assessment (EQA) or external proficiency testing is an essential component of assessment for BEs *(3,4)*. The standards of performance in the laboratory and the reliability of test results are monitored and improved by periodically evaluating performance against an established and independent quality assessment panel. Box 9.2 summarizes the benefits of EQA.

EQA involves the provision of panels of test samples to participants for inclusion in their routine testing operations. The contents of the analytes in test samples are unknown to the participants but known to the EQA provider. Results are returned by each participant to the EQA provider and analysed with the results from all other participants. This allows comparison of a laboratory's performance with that of other laboratories that have tested the same samples and/or with results from a reference laboratory *(17,18,59)*. Errors identified by EQA are analysed so that CAs and PAs can be taken; when performed properly, this helps to minimize errors.

Box 9.2 Benefits of EQA *(12,15,17,18,20,59,74)***:**

- enables interlaboratory comparison;
- encourages exchange of information between laboratories about test kits, equipment and methods;
- provides opportunities for education and training or retraining of staff, and increasing staff confidence;
- help to minimize errors and non-conformances;
- facilitates self-assessment and audits, and supports the accreditation process; and
- enhances confidence among a BE's stakeholders and customers.

9.2.1 Participation in EQA schemes (EQAS)

BEs should participate in EQAS for all tests they perform *(15,17,18)*, particularly those that affect blood product and patient safety.

BEs can participate in EQAS that are commercially provided or that are organized at international or national level. BEs planning to participate in EQAS should explore what is available and select the most appropriate based on their needs and resources. One important consideration is the ability of the EQA provider to ship the samples to the BE in a manner that ensures sample integrity and preservation of the analyte to be tested. If the EQA provider is based overseas, the BE should always take note of any border controls that may affect timely delivery of the samples and make the necessary arrangements to mitigate the risk of delays. The key steps in BEs' participation in EQAS are described in Table 9.2.

Table 9.2 Key steps in participation in external quality assessment schemes (EQAS)

Step	Activity
1	Register with the selected EQAS programme. Check on any border or import controls that may require special shipping arrangements or permits.
2	Train the staff who will be participating in the EQAS.
3	Upon receipt of the EQAS materials, check that there has been no tampering or deterioration of samples (e.g. haemolysis) during shipment. Store the materials as instructed.
4	Review the instructions for use or information manual accompanying the EQAS materials, including the reporting format or documentation.
5	Conduct the testing on the EQAS materials as follows: • Use the same test kits and technology as used in routine testing of samples. • Assign the same staff who conduct routine testing of samples to test the EQAS samples (not senior or more experienced staff). • Test the samples alongside blood donor/patient samples. • Use the same approved algorithm as is used for routine samples. • Test the samples on all instruments that follow the same algorithm. If the quantity of sample is insufficient, the instruments should be rotated to ensure that all the instruments are included in EQAS and perform in the same manner.
6	Encourage all the staff who are participating in the programme.
7	Record the results. Check that the recorded results are accurate and follow the instructions to avoid clerical errors (which do happen even with EQAS samples!)
8	Return the results to the organizer before the closing date.
9	Review the EQAS report and take corrective action in response to any non-conformances or unsatisfactory findings.
10	Share the results with the staff. Obtain permission from the advisory committee before publication of the data, if necessary.
11	Define the indicators used to assess any improvement as a result of EQAS implementation.

WHO encourages the establishment of EQAS via the BE, the national health authority or other qualified organization at the national, regional, provincial or state levels according to the needs of the country *(12,15,17,18)*. Some BEs may decide to establish their own EQAS in their country. BEs planning to organize EQAS may find further information about establishing and operating EQA programmes in the WHO guidelines for establishing EQA in screening for TTIA and for blood group serology *(17,18)*.

9.3 QIs

QIs are measures that give an indication of process, procedure or output quality. QIs are useful quality management tools for monitoring and evaluation (M&E) (Box 9.3). Whereas monitoring involves the ongoing collection of data, evaluation is the systematic and objective review of data as a basis for making decisions and taking action. M&E helps to improve performance and achieve better results by establishing links between past, present and future actions.

Box 9.3 Benefits of quality indicators

- important tools to monitor quality improvement in the BE;
- provide input for management review;
- help to measure BE performance and benchmark it against that of other institutions with similar characteristics;
- provide insight into level of quality of product and services and their trends over time;
- identify weak links in processes and prioritize issues to be resolved;
- contribute to assessment of efficacy or corrective measures;
- assess quality system performance in meeting set goals;
- prerequisite for accreditation or certifications.

QIs may be classified as internal or external:

- **Internal QIs** are defined and used by management to control their own processes, improve quality and achieve better management results. Internal QIs are detailed, specific and address problems of local interest. For example: venepuncture failures, clots in red blood cell concentrates or expired platelet concentrates.

- **External QIs** are global and general, and enable surveillance of the indicators by external partners, for example, regulatory authorities. They could include number of donations collected per 1000 inhabitants and percentage of donations collected from first-time donors.

9.3.1 Selection and use of QIs

QIs should be defined, monitored, collated, analysed, reported and implemented for all processes in the BE (Box 9.4). The ISBT working party on quality management compiled a list of commonly monitored QIs after conducting a survey among BEs *(75)*. The role, importance and procedure for implementing and monitoring QIs in BEs has also been reviewed *(76)*.

Box 9.4 Desirable characteristics of selected QIs

- Qis should be objective, feasible, reliable and measurable indicators.
- They should measure the processes and their outcomes in the BE.
- They are usually expressed as absolute numbers or percentages; if percentages or proportions are used, make sure the numerator (observations of interest) and denominator (total number of observations) are defined.
- Select QIs based on institutional priorities, criticality and frequency of a particular problem, evidence and expert consensus.
- The number of indicators will depend on the number, extent and nature of activities performed.
- Focus on critical indicators and cover all key areas of activity in the BE.
- Results generated from QI data should be reliable, easy to interpret and comparable.
- Clearly define and document the methods for obtaining, calculating and expressing each QI (including units of measurement).
- Define the source of data, person(s) responsible for data collection and analysis, and timelines for data analysis.

Organizational QIs are dynamic and can be changed by management decision when there is a change in organizational needs and priorities, or a system or process change. Results generated from QI data should form the basis for implementation of CA and quality improvement. Periodic reports should be submitted to the management and other regulatory bodies and authorities concerned. These should also be easily accessible by BE staff. BEs are generally obliged to notify sentinel events to the national authorities.

9.4 Data management

Data quality management is a set of practices undertaken to maintain high-quality information throughout the process of handling data; from acquisition, to implementation, distribution and analysis. Tools for data collection and analysis include the charts, maps, and diagrams designed to collect, interpret and present data for a wide range of applications including quality management. The use of dashboard tools also helps to provide valuable insights when monitoring data trends.

High-quality data are necessary to support decision-making activities at all levels. Data management includes gathering and analysis of accurate and appropriate data to determine if the quality system is suitable and effective and identify opportunities for improvement. By using data from well-designed monitoring and measurement activities, the BE can decide which areas to focus its improvement activities on. This enables more efficient deployment of limited resources to achieve optimal impact in improving overall performance and customer satisfaction.

For maximum effectiveness, the results of data analysis must be communicated in a timely manner. Accuracy, consistency, validity, completeness, integrity, timeliness and uniqueness are important prerequisites for data metrics. Sources of data in the BE include record books, forms and computer reports. Responsible personnel who are assigned to collect and analyse the data should be properly identified and defined, and these tasks should be considered as part of their work assignments. Appropriate statistical methods should be employed during data analysis and should be selected depending on the information required from the dataset.

Descriptive (or diagnostic) analysis is useful when there is a need to study what has happened and why. It usually starts with identifying the variables that will provide a representative description of the dataset. The results of descriptive analysis are usually communicated through appropriate tables, charts and histograms. It is always important when reviewing these results to bear in mind how the data were collected, whether the sample size was adequate, any specific factors that may have influenced the results and whether the data have been presented appropriately.

Predictive analytics are useful in answering questions that relate to the future. This generally requires more advanced statistical calculations and interpretation, and a clear understanding of uncertainty limits in the results.

Once the results of data analysis are obtained and verified, they should be shared with relevant bodies. In the case of data such as QIs, this may include BE staff and regulatory authorities. When sharing data, measures should always be in place to ensure data security and to maintain confidentiality of sensitive donor, patient, organizational and staff information.

Further information regarding the collection, analysis and use of data can be obtained from relevant textbooks and materials about data science and statistics. It is important to be aware of the factors and considerations that can affect or influence the validity of conclusions derived from the data analysis. Misleading or incorrect conclusions from poorly collected data and improperly applied data analysis are worse than having no data at all.

9.5 Haemovigilance

Haemovigilance is defined as the set of surveillance procedures covering the entire blood transfusion chain, from the donation and processing of blood and its components, through to their provision and transfusion to patients, and including follow-up *(77)*. Thus, haemovigilance includes the monitoring, reporting, investigation and analysis of adverse events related to the donation, processing and transfusion of blood, and actions taken to prevent their occurrence or recurrence. Adverse events include all reactions, incidents, near-misses, errors, deviations from SOPs and accidents associated with transfusion and blood donation.

This information plays a fundamental role in enhancing patient and donor safety. Learning from adverse events and identifying systematic problems can drive the introduction of measures to enhance the quality, safety, efficacy and cost-effectiveness of blood and blood products as well as the donation and transfusion processes.

Haemovigilance should be fully integrated into the quality system of the BE and include *(77)*:

- donor haemovigilance, including recognition and clinical management of adverse events associated with donation, and their monitoring, reporting, investigation and analysis;
- implementation of policies, guidelines, protocols and SOPs for all processes involved in the donation and provision of blood and blood products;
- epidemiological surveillance of donors, post-donation information and look-back;
- identification, recording and reporting of near-misses, errors and deviations associated with these processes, and abnormalities in intermediate and finished blood products;
- traceability of each donation from the donor to blood processing, the blood product, its issue to a health care facility and transfusion to the patient, and vice versa;

- responding to notification of an adverse event in a patient by retrieving all blood components associated with the index blood component(s);
- implementation of haemovigilance as part of the quality system, and mechanisms for taking CAs and PAs and monitoring their outcomes;
- training and assessment of staff involved in all steps along the blood transfusion chain (collection, processing and provision of blood and blood products);
- liaison with hospitals: administration, blood banks, transfusion committees and clinical services.

More detailed information on haemovigilance systems, including haemovigilance practices at hospital level, is provided in the WHO guidance documents on haemovigilance *(5,77)* and other publications *(78,79)*. WHO has also produced a user guide for navigating resources on haemovigilance systems to assist countries in developing and strengthening their own haemovigilance systems *(80)*.

9.5.1 Traceability

Traceability is the ability to link the donor, donation and its associated components with the ultimate recipient. This can be extended to include all devices and consumables used in the collection, processing and testing of the donation. Box 9.5 describes key measures to ensure traceability.

Bidirectional traceability – the link from donor to recipient and vice versa – is a fundamental prerequisite for establishment of a haemovigilance system and must be ensured "from vein-to-vein". This enables investigations to be conducted for routine audits and when things go wrong – for example, rechecking the donor's testing results after a report of suspected TTI, investigating a suspected recipient transfusion reaction, recall of blood products or look-back studies.

Box 9.5 Measures to ensure traceability

- Assign unique identifiers to each donation event and each component.
- Maintain records of processing and testing.
- Retain data linking a particular component to the donor and collection (date).
- For each donation or component, record details of the transfusing facility to which it was distributed, or its disposition (e.g. damaged, discarded, outdated).
- Ensure that hospitals record and retain the details of components issued and transfused, as well as relevant patient information.
- Ensure that hospitals and other facilities receiving components retain the records of components taken out of stock (e.g. on expiry) and discarded.

9.5.2 National haemovigilance systems

Haemovigilance may operate at local and institutional levels, but national coordination and management are vital for effective surveillance. An effective haemovigilance system requires coordination and collaboration between multiple stakeholders involved in the collection, provision, transfusion, surveillance and regulation of blood and blood products. Haemovigilance should be incorporated into the national blood policy *(77)*, and a national haemovigilance system established to enable continuous quality improvement of the transfusion chain, enhance donor and patient safety, and reduce wastage. Further information on national haemovigilance systems is provided in the WHO guide to establishing a national haemovigilance system *(5)*.

9.6 National licensing

Licensing (or authorization) is usually granted by the health authority to the BE to carry out the activities of collecting, testing, processing, storage and distribution of human blood intended for transfusion as whole blood or its components. This usually indicates that the health authority has approved the BE's operations following a thorough evaluation of its compliance with GMP and other recognized standards of best practices in BEs. This process may involve various agencies or departments that have health responsibilities. These agencies may carry out visits to evaluate or verify compliance with the applicable conditions and may conduct audits or inspections related to the granting or the ongoing retention of licensing, certification of capacity or authorization of a BE. The authorization process generally has a defined period of validity during which the BE must continue to fulfil the conditions that allowed its authorization. Governmental agencies such as the ministry of health and NRAs are the most common licensing bodies *(81)*.

9.7 Accreditation programmes

Accreditation is a nongovernmental, voluntary process that evaluates institutions, agencies and educational programmes. It is both a process and a credential. The accreditation process is voluntary and should be conducted by external parties. Only organizations, agencies or programmes can be accredited. If an organization is accredited, this means a thorough assessment was conducted against recognized standards of best practice and that the organization was able to demonstrate evidence that these standards are being met. It is usually a rigorous process conducted by a third-party organization. The accreditation process is typically repeated every 2–4 years, depending on the accrediting body *(82)*.

Accreditation is not the same as licensing, although some regulatory bodies may require accreditation as part of the licensing process. A BE may choose voluntary accreditation of its activities through various organizations, some of which provide full accreditation for the entire range of BE activities. Examples of the latter include the AABB, AfSBT and PAHO.

CHAPTER 10

Steps in implementing an effective quality system in the BE

There is no universally accepted process for implementing a quality system in a BE. The order and manner in which elements of the system are introduced depends on the local priorities, circumstances and available resources. It is a dynamic process involving many interconnected parts, and a systematic approach will help identify key steps and potential barriers. A stepwise approach will allow smoother implementation and facilitate engagement of all key stakeholders. A quality culture as well as stewardship and ownership of the quality system are important prerequisites.

10.1 Practical considerations

10.1.1 Relevance in the context of the BE and the country

When planning a quality system that is appropriate in the context of the country, consider the following aspects:

- the architecture of the blood transfusion chain in the country and any existing quality activities, including national, regional, local and institutional participants and arrangements;
- oversight and governance responsibilities;
- existing and potential resources, including staffing and funding; and
- education and training needs and resources.

An effective quality system should enable BEs to demonstrate that all steps in the transfusion chain have been carried out according to well-characterized and established standards (Box 10.1).

Box 10.1 Examples of established standards for steps in the transfusion chain

- Donors have met the standard eligibility criteria and were healthy at the time of the phlebotomy.
- The processing of the whole blood collected to obtain the various blood components was performed according to validated methods to generate products of suitable quality.
- The screening and testing in place will detect agents potentially involved in transfusion-transmitted infections and therefore ensure the safety of the supply of blood.
- Storage and distribution are organized in a way that is not detrimental to the quality of the blood and blood products.
- Measures are in place to reduce the recurrence of unexpected reactions in donors or recipients.
- The continuous improvement of the quality system is supported by periodic revisions.

10.1.2 Planning and prioritizing according to available resources

Following the decision to implement a quality system, the BE should be realistic in evaluating the resources available and not underestimate the tasks that need to be performed to ensure a successful outcome. Appropriate funding should be available not only to implement but also to maintain and ensure continuous improvement of the quality system over time. A sufficient number of staff with relevant qualifications and skills should be present, or available to be hired at short notice. The facilities should have the necessary infrastructure, with areas organized in a logical sequence for confidential donor interviews, blood collection, blood processing, blood testing, component storage and waste disposal. These requirements should also be met by mobile blood collection sites and the related blood transportation activities. Equipment and tools necessary to carry out all related activities should be in place or purchased. Information technology should be used to support the implementation and maintenance of the quality system, if possible, and qualified staff, hardware and software should be available on site or on call.

Timelines for implementation should be realistic and achievable. If any of the resources is identified as a limiting factor, the BE should adjust its priorities accordingly and consider a stepwise implementation of the quality system, starting from a basic system and growing over time to a system fully compliant with international guidelines and/or regulatory requirements.

Organizations such as the AfSBT follow a stepwise programme to accreditation that takes into account that BEs operate at different levels of development *(83)*.

10.2 Gap analysis and development of an action plan

10.2.1 Gap analysis

To implement a quality system that will meet the needs of the BE, it is important to first identify, document and understand all applicable requirements, including regulatory ones. These may be organizational (for example, quality representative, independent QA function) or process-related, and must include national regulatory requirements.

An assessment of the current situation in the BE against the identified requirements is then conducted to identify gaps and opportunities for improvement. The assessment can include interviews with staff, direct observation of operations, and a review of policies, processes, procedures, records and practices. BEs may also consider using consultants with expertise in GMP-based quality systems as independent assessors.

10.2.2 Development of an action plan

Once the gaps and opportunities for improvement have been identified, an action plan to address and close them can be developed based on the results of the gap analysis. The action plan should prioritize activities depending on the nature of the gap and the risk associated with each of the gaps. The roadmap should identify the activities and resources required; competencies of staff; responsibilities and authorities; and timelines (time frame).

Specific guidance on drawing up an action plan can be found in the section on quality management system development and implementation of the roadmap in the *WHO guideline on the implementation of quality management systems for NRAs (84)*. The information presented in this document is easy to adapt and can be applied in developing the action plan for BEs.

Establishing the BE's documentation system to manage the new policies, SOPs and records is always an important first step.

10.3 Steps in implementing a quality system

Potential mechanisms that can support quality system implementation include establishing effective coordination and communication channels and implementing applicable information and communication technologies tools for internal and external application of the quality system, including communication of quality policy awareness. Engaging with customers and stakeholders is also helpful. Other supportive mechanisms include the creation and implementation of training plans for quality staff and granting assigned quality system staff the authority and responsibility to implement and maintain the system. Crucial factors for success are support from senior management and establishing high-level ownership, stewardship and commitment to the implementation and maintenance of the quality system.

Quality system implementation roadmaps should be included in the BE's strategic plans by senior management and in the national health strategic plans. Responsibilities and authorities for contributing to the quality system should be included in every staff job description and performance appraisal. Tables 10.1 and 10.2 suggest the steps to be taken by BEs when developing and implementing a quality system and related procedures.

Table 10.1 Suggested steps in the development and implementation of a quality system

Step	Activity	Responsibility
1	Assign resources: staff, financial, equipment and infrastructure. This includes appointment of a quality manager followed by designation of quality leads in each area.	Top management
2	Use results from the gap analysis to determine the status of the quality system and submit a report to top management, noting activities and areas that require actions.	Assigned staff, QM/ consultant
3	Prioritize activities based on: • availability of resources, internal and external • risks of non-implementation of quality system • regulatory requirements of national laws and regulations.	Top management, QM
4	Allocate responsibilities and authorities with timelines for development, review, approval, implementation, and monitoring and evaluation of prioritized quality system requirements.	Top management, QM, assigned staff/ consultant
5	Collect input and feedback from staff in the relevant departments to promote ownership of quality system implementation.	Top management, QM, assigned staff/ consultant

Table 10.1 *continued*

Step	Activity	Responsibility
6	Consolidate the feedback and input into an action plan, for use as a roadmap for quality system implementation.	QM, assigned staff/ consultant
7	Integrate the quality system action plan/roadmap into the blood establishment's organizational action plans and strategic plans. Integrate into the ministry of health's strategic plan or policy, where possible.	Top management
8	Develop and document SMART quality objectives (specific, measurable, achievable, realistic, timely), including a plan for monitoring and evaluation with the required resources.	Top management, QM, assigned staff/ consultant

QM, quality manager.

In all instances, the BE should either develop new procedures or harmonize with existing procedures. In practice, work will need to be done simultaneously on several of the steps shown in Table 10.2. The process should continue until all procedures in each area are complete. Top management, the QM and section leaders will need to be involved in all steps. Overriding policies will also need to be developed for all elements of the quality system, including a quality policy.

Table 10.2 Suggested steps in development of quality system procedures

Step	Quality system element	Activity
1	Documentation	• Determine documentation system structure; design templates for standard operating procedures (SOPs) and forms. • Develop a procedure for document control. • Train all staff on system requirements, and SOP writers and form designers on technique. • Designate an officer responsible for the document system and document control.
2	SOPs	• List all processes in each technical and administrative area. • Draw up a list of anticipated SOPs required in each area; update the list as progress is made. • Prioritize and map key processes; start developing SOPs and related forms.
3	Training	• Develop procedures for staff training, communication and determination of competency. • Provide training on each element of the quality system as procedures are developed. • Develop procedures for handling organizational charts, job descriptions and staff records.
4	Equipment	• Compile a list of all equipment in each area ensuring that each item has a unique identifier. • Develop procedures for equipment evaluation, selection, operation, validation and calibration. • Develop procedures and schedules for maintenance and cleaning of equipment.
5	Audits	• Develop procedures for internal audits together with checklists and a report template. • Select and train internal auditors from all areas; monitor their performance over time. • Schedule and perform audits in all areas. • Develop procedures for required external audits, including audits of blood establishment suppliers.
6	Error management	• Develop procedures for error management and improvements and put them into practice. • Design a standardized template for error reporting. • Monitor trends over time to identify improvements.
7	Customer feedback	• Develop procedures for monitoring customer satisfaction and handling of complaints and put them into practice. • Design forms and reporting mechanisms; monitor trends.

Table 10.2 *continued*

Step	Quality system element	Activity
8	Review of data	• Develop procedures for organizational data management and change control. • Decide on quality indicators to be monitored; design reports and disseminate data. • Develop procedures for haemovigilance, participation in external quality assessment schemes and performance of reviews such as quality reviews and product quality reviews.
9	Risk management	• Develop procedures for risk management and contingency planning and put them into practice. • Communicate procedures and plans to staff and conduct practice runs.
10	Management review	• Develop procedures for management review; decide on inputs and outputs required. • Schedule regular reviews and mechanisms for reporting back. • Modify and improve quality system and operations based on the results of the reviews.
11	Quality manual	• Develop and finalize the quality manual covering all elements of the quality system. • Compile a master list of all documents and keep it up to date.

The plan for designing and developing a quality system should include logical and realistic short-, medium- and long-term strategies based on a stepwise approach (Box 10.2). A timeline with milestones should be drawn up and should reflect a sequential approach. This could be represented in a Gantt chart, in which a series of horizontal lines shows the amount of work done in certain periods of time in relation to the work planned for those periods.

Principles of GMP should be incorporated into the quality system to assure the safety and quality of blood and blood components. Having GMP in place ensures that blood and blood components are consistently produced and controlled to meet predefined quality standards appropriate for their intended use. The WHO and PIC/S *Good practice guidelines for blood establishments and hospital blood banks (4,71)* are good sources of information. Accreditation of the organization may also be considered once the quality system is established.

Box 10.2 Helpful tips for implementing a quality system

- It may be useful to appoint a steering committee.
- Create a sense of excitement about the project with the whole BE working towards a common goal.
- Develop quality champions/teams in each work area.
- Initiate work on several elements of the quality system at the same time; do not wait to finish one before starting the next.
- Draw up separate action plans for each element of the quality system to help track progress.
- Tackle tasks methodically and step by step so as not to feel overwhelmed.
- Hold regular meetings with the staff involved to discuss progress and resolve any problems.
- Keep top management informed.
- Adopt tight but realistic schedules.
- Adapt and update action plans to keep pace with changing circumstances.
- Recognize staff efforts and the achievement of important milestones.
- Remember, a quality system is never done and will need ongoing improvements to keep it fresh.

All interconnected parts of the quality system should be reviewed and updated regularly to ensure that it is effective. The Deming cycle of "plan, do, check, act" (PDCA) is a central concept in the continuous improvement of quality management but can also be applied in the implementation of a new quality system to help resolve any identified issues. It is also useful for the ongoing improvement of quality activities and is explained in Chapter 11.

Annex 1 contains a summary checklist for implementing a quality system in a BE. This checklist may be useful for monitoring the progress made during the implementation of a quality system.

10.4 Integration with national quality policies and strategies

It is important that the BE takes account of prevailing national blood policy, strategies and plans as well as relevant national laws and regulations when developing their own quality system. At the national level, it is essential that the important role of quality systems in BEs is included in the national blood policy and national strategies implemented to achieve effective and functional quality systems in BEs.

Such strategies should include *(4,21)*:

- national standards on GMP for BEs;
- national guidance on operational systems and processes for blood services;
- national standards for blood and blood product quality and safety;
- national guidelines on the appropriate clinical use of blood;
- national guidelines on procurement practices for equipment, consumables and reagents;
- a national framework for the quality system;
- information networks to facilitate quality practices and compliance with quality standards; and
- appropriate regulatory frameworks that incorporate requirements for a quality system in BEs.

CHAPTER 11

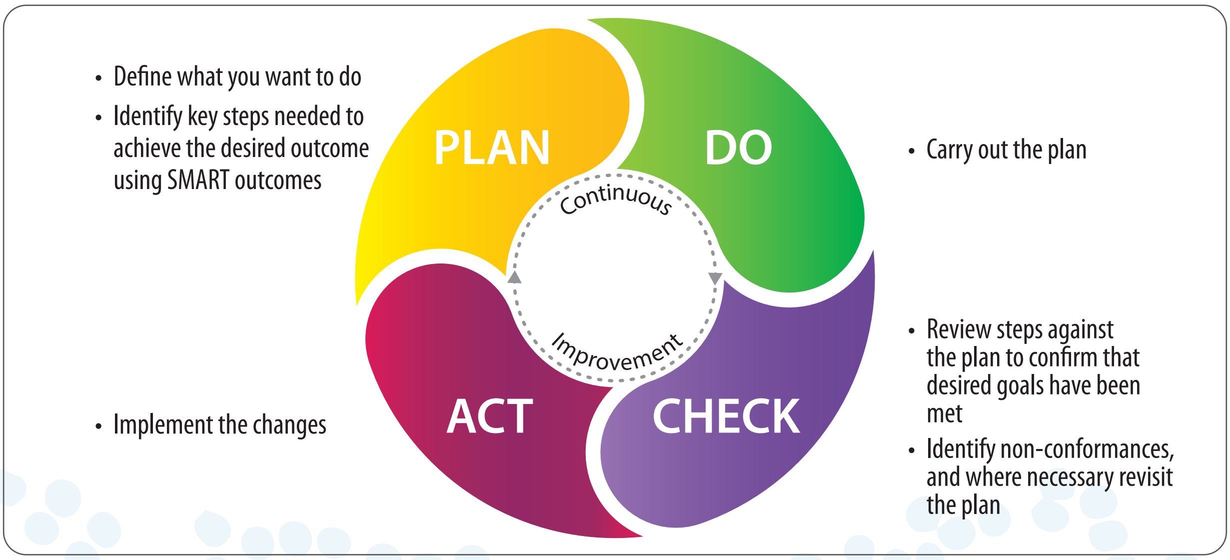
Continuous quality improvement

Maintaining quality requires continuous improvement. It is an ongoing process of seeking better ways of doing things and ensuring that everyone is engaged in quality management. Quality management focuses on people, procedures, documentation and evidence to review and track performance throughout the vein-to-vein chain. Achieving the level of quality management necessary to assure the safety and efficacy of blood products for transfusion is an intensive and stepwise, but continuous, journey that, in the long run, avoids unnecessary costs but above all ensures the well-being of patients and blood donors.

11.1 The quality cycle

All organizations change over time. Quality systems play a key role in ensuring that change is undertaken in a controlled way and that the desired outcomes are achieved. The "quality cycle" is a method first described by Edward E. Deming in 1950. The PDCA version is commonly used to manage change. The outline of the cycle is shown in Fig 11.1.

Fig. 11.1 The Deming cycle: plan, do, check, act



Source: adapted from *WHO user guide for navigating resources on stepwise implementation of haemovigilance systems (80).*

The PDCA cycle is an iterative, four-stage approach for continually improving processes, products or services and for resolving any issues identified. It involves the systematic development of change plans (for example, new equipment qualification, new blood component production) and their implementation. This ensures that all requirements identified have been met prior to the introduction of the new process, product or service into the production environment.

This practical PDCA framework and structure for identifying improvement opportunities and evaluating them objectively can be applied to all BE activities, both during implementation of a new system and during ongoing quality improvement activities (see Table 11.1). Deming's approach is not just about improving individual processes in the vein-to-vein chain, but about improving the whole system.

Table 11.1 Elements of the "plan-do-check-act" (PDCA) cycle

Element	Activity
1. Plan	The planning phase of a quality improvement initiative involves preparing an outline of any change process, identification of desired outcomes and documentation of the steps needed to achieve the goals. • In the context of new equipment or processes, this might be documented in a **change control plan**. • For an initiative relating to clinical transfusion practice, this phase might involve identification of the standards being assessed, the datasets that will be used to determine if the standard is being met, and an outline of the processes and people needed to carry out the study.
2. Do	Carry out the plan.
3. Check	Collect evidence to demonstrate whether the goals of the change process have been achieved and, where appropriate, document changes to the plan or non-conformances.
4. Act	Make the change.

Ideally any intended outcomes identified in the planning phase should be SMART in nature. SMART goals are:

- **S**pecific – identify the desired outcome of the change;
- **M**easurable – identify the way in which success will be determined;
- **A**chievable – relevant to the current status of the organization, its resources and its personnel;
- **R**elevant or realistic – assess that the goals should/can be achieved; and
- **T**ime bound – clearly documented dates for completion of each phase.

Some authorities extend the SMART approach to the use of SMARTER goals. The additional requirements involve **E**valuation and **R**evision. Evaluation refers to ongoing review of outcomes to demonstrate that these are sustained over time and revision provides a mechanism to review or revise the goals or plan when change has not been successful.

The PDCA cycle also functions as a tool for systematic improvement of processes and products using data collected during their use or production. This might involve QC data or information on incidents collated within the wider quality system.

11.2 Tools for continuous improvement

Several approaches that can be taken to improve performance and quality within an organization, including the use of Lean and Six Sigma methodologies. Six Sigma is a business methodology for quality improvement that counts how many defects there are in a current process and seeks to systematically reduce the probability that an error or defect will occur. It uses qualitative and quantitative techniques to drive process improvement. Lean management is an approach to managing an organization that systematically seeks to achieve small, incremental changes in processes that will lead to improved efficiency and quality.

The five principles of Lean and Six Sigma are:

- focus on the attributes most important to the customer;
- assess the value chain and find the problem;
- remove waste (including non-value-added steps);
- communicate with your team, involve stakeholders; and
- create a culture of change and flexibility.

The key to both approaches is a focus on reviewing systems and processes using data, and engaging staff in the process. Initial progress towards developing a positive quality culture can, however, be achieved without these types of systems. There are a number of practical ways by which staff engagement can be improved, including:

- consulting with staff and providing opportunities for them to review and comment on initiatives that will have an impact on them and on the wider organization;
- providing opportunities for cross-departmental discussion and problem-solving;
- regularly informing staff about progress towards meeting targets, QC data and incidents; and
- empowering staff to make suggestions for change and actively involving them in projects and initiatives.

11.3 Plasma fractionation and quality system improvement

The introduction of plasma fractionation to produce PDMPs from domestic plasma is a good example of the way in which change can lead to strengthening and improvement of quality system in the BE. In this situation, the BEs involved are required to ensure that the plasma for fractionation meets required quality standards. Plasma for fractionation is normally regarded as a raw material for the fractionation process. The quality of the plasma is therefore a major concern during fractionation, and this applies both to fractionation by a national fractionator and to contract fractionation used to produce plasma products. Quality requirements for plasma for fractionation are additional to those for standard blood components.

The BE normally has to comply with GMP requirements. The implementation of GMP during blood and plasma collection procedures helps to assure consistency in applying donor selection criteria for each donation, prevent errors during testing and support traceability by adequate documentation of all relevant steps. The EU Guidelines to GMP require that "the rules designed to guarantee the quality, safety and efficacy of medicinal products derived from human blood or human plasma must be applied in the same manner to both public and private establishments, and to blood and plasma imported from third world countries". In general, within the setting of contract fractionation, no distinction is made between plasma collected for manufacture of PDMPs in the country where the fractionation facility is situated and that for products that will be returned only to the original country. Fractionators will approve individual BEs for the supply of plasma for fractionation and will normally undertake a quality audit of the site as part of the approval process.

As part of GMP requirements, a plasma master file (PMF) must be prepared by the fractionator *(85,86)*. The PMF documents technical information relating to the collection, testing and processing of blood and plasma, such as collection sites, donor selection criteria, testing sites, materials used in collection of blood (including blood bag types), equipment and materials used for testing of donations, and the logistics systems used to transport plasma. The PMF also includes epidemiological data on transfusion-transmissible infections. The PMF is compiled with the quality information provided by the BE and updated annually for submission by the fractionator to the relevant regulatory authorities. The BE quality department plays an important role in collating and maintaining the data for the PMF.

Some blood services may decide to centralize their blood donation processing and testing functions to facilitate the production of plasma for fractionation *(38)*. This will involve significant change in the operations of the BEs undergoing centralization. Movement of blood donations and blood components across several sites will require adoption of common standards across the network. Processes will need to be developed for transportation of blood samples and blood components, materials and information between sites. Implementation of computerized information systems and possibly systems for high throughput screening of blood donations will also be required. Management of change associated with this type of development will require investment in the quality system and the quality team. This might involve the need to develop:

- GMP-compliant contracts between the BEs involved in the centralized network and between external suppliers and the BEs;
- interfacing of automated testing platforms and the blood management system to facilitate automated transfer of test results leading to a reduction in the risk of transcription errors;
- validation of automated systems for blood donation testing and qualification of individual testing platforms, and ongoing monitoring of test results and participation in EQAS;
- labelling all blood donations, laboratory samples and blood components with unique identification codes that permit bidirectional traceability, including specific codes for each BE within the network;
- introduction of software for monitoring the dispatch and receipt of blood and blood products and laboratory samples, including information on mode and conditions, and times of shipment and arrival.

Introduction of fractionation of plasma will result in progress and improvement at different levels: use of domestic recovered plasma (instead of discarding it), stringent compliance with GMP for the BE, standardization of practices in participating BEs and, finally, access to needed PDMPs.

To support such initiatives, the WHO Achilles project *(19)* was implemented to assist countries in improving the quality of plasma suitable for fractionation, and this includes working with regulatory authorities and national blood services to implement blood regulatory systems that will help to strengthen quality systems in BEs.

11.4 The need for continuous improvement of the quality system

All blood services change over time and develop in parallel with the development of the wider health system in which they exist. BEs usually develop in a stepwise manner, largely driven by changes in demand from user hospitals. Over time hospitals will require access to a wider range of, often more complex, blood products. The BEs, therefore, are likely to require access to better and more complex technologies and skills to deliver the more complex products. An initial focus on provision of whole blood for use in major bleeding events is followed by the introduction of blood component therapy and then by use of surplus plasma for fractionation. Consolidation of BE activities to increase throughput may involve centralization of testing and processing of blood donations and this is often associated with the implementation of newer and more complex technologies. Quality plays an important role in supporting changes of this type and ensuring that change is successfully delivered. The quality system will need to improve progressively to achieve this.

The *WHO Aide-Memoire for National Blood Programmes (3)* identifies the need for development of necessary legislation and/or regulation for the blood services. The establishment of a national regulatory authority will increase the quality expectations of BEs. This will necessitate quality inspections based on GMP and greater scrutiny of the rationale for new products, processes and the plans for their implementation. Stepwise improvement in the quality system will be needed to ensure that these expectations are met.

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Annex I
Summary checklist for implementing a quality system in a blood establishment (BE)

Chapter no.	Quality system element	✔ Requirements
3	Resources	❑ Sufficient, qualified, trained and competent staff ❑ Physician responsible for medical matters including safety of donors and patients ❑ Premises suitable for workflow of BE ❑ Suitable, properly functioning, well-maintained equipment ❑ Adequate supplies of reagents and consumables from approved suppliers ❑ Sufficient financial resources for continuity of operations ❑ Financial strategies, plans and budgets ❑ Occupational health and safety programme, provision of personal protective equipment ❑ Risk assessment, mitigation and preparedness with contingency plan vBusiness continuity planning
4	Organizational management	❑ Senior management commitment and support for quality system ❑ Quality manager appointed and trained, competent quality staff ❑ Organizational chart showing hierarchical structure of BE ❑ Individual organograms for each work area ❑ Job descriptions for each position ❑ A culture of quality ❑ Process maps with identified critical control points for key processes
5	Standards	❑ Appropriate national or international BE standards ❑ Access to relevant guidelines and guidance documents ❑ Biological Reference Standards used in laboratories
6	Documentation	❑ Quality policy aligns with BE's vision, mission and strategic direction ❑ Policies define BE's overall intentions and direction ❑ Quality manual and master list of documents ❑ Standard operating procedures for each technical and administrative area ❑ Forms, paper-based or electronic, for collection of data ❑ Traceability via records ❑ Document control system for organizing, routing, tracking, authorizing, distributing, archiving and destroying documentation
7	Training and education	❑ Initial training for new staff followed by ongoing training/education ❑ Assessment of training needs ❑ Training policy supports BE's mission and objectives ❑ Training activities planned to ensure training goals are met ❑ BE staff trained in principles of quality, the quality system, documentation and use of quality monitoring tools ❑ Continuing education programmes for staff ❑ Education programmes for health-care professionals involved in blood transfusion facilitated or supported ❑ Post-training evaluation

Chapter no.	Quality system element	✔ Requirements
8.1	Validation	❑ Evaluation to select most suitable equipment and reagents ❑ Validation of processes, methods, equipment and reagents ❑ Approved validation plans/protocols ❑ Validation sample size sufficiently large for test of statistical significance ❑ Formal reports prepared after each validation
8.2	Change control	❑ System, policy for change control ❑ Post-implementation evaluation
8.3	Error management, corrective and preventive action	❑ Error management process covers errors, incidents and near-misses ❑ Encompasses corrective action and preventive action ❑ Includes error detection, reporting, initial evaluation, remedial action, in-depth analysis of root cause, corrective and preventive actions and evaluation of effectiveness of actions taken ❑ Errors are systematically categorized and analysed; trends are monitored
8.4	Management review	❑ Periodic management reviews with clear performance indicators ❑ Results of reviews communicated to staff ❑ Actions implemented according to agreed timescale ❑ Updating of quality policy and quality objectives where indicated
8.5	Product quality review (PQR)	❑ Regular PQRs in all facets of blood transfusion chain ❑ Trends highlighted and opportunities for improvements identified
9.1	Internal and external audits	❑ Periodic internal audits conducted by trained staff from different departments ❑ Audit plans compiled in advance; checklists drawn up ❑ Audit reports disseminated ❑ Corrective actions completed within agreed time frames ❑ Actions taken reviewed for effectiveness; trends monitored ❑ External audits, including supplier audits, performed
9.2	External quality assessment schemes (EQAS)	❑ Participation in EQAS for tests affecting blood product and patient safety ❑ Participating staff trained ❑ Analysis of errors identified; corrective and preventive actions taken
9.3	Quality indicators (QIs)	❑ QIs defined, monitored, collated, analysed and reported vPeriodic QI reports sent to management/regulatory authorities
9.4	Data collection and analysis	❑ Data collection and analysis tools used (charts, maps, diagrams) ❑ Data processed; results communicated in a timely manner
9.5	Haemovigilance	❑ Haemovigilance system integrated into quality system ❑ Staff trained in haemovigilance across all steps in transfusion chain ❑ Liaison with hospitals, transfusion committees, clinical services etc.
9.6	National assessment	❑ Authorization/licence granted by national health authority ❑ System for audits or inspections
9.7	Accreditation	❑ Accreditation process (voluntary unless part of licensing agreement)

Chapter no.	Quality system element	✔ Requirements
10	Quality system implementation	❑ Gap analysis performed ❑ Action plan developed for stepwise approach ❑ Good Manufacturing Practice (GMP) principles incorporated into quality system
11.1	The quality cycle	❑ Deming PDCA (plan-do-check-act) cycle used for continually improving processes, products and services (optional)
11.2	Plasma for fractionation	❑ Compliance with GMP requirements if plasma is supplied for fractionation ❑ Information provided by BE for inclusion in plasma master file ❑ Quality audits by fractionator
11.2	Lean, Six Sigma	❑ Lean and/or Six Sigma approaches followed to enhance continuous improvement (optional)

Implementation of a quality system is a complex and dynamic process involving many interconnected parts, and a systematic approach will help the blood establishment (BE) to identify and prioritize key steps. This checklist may be useful for monitoring the progress made during the implementation of a quality system. BEs may also wish to use the information in the checklist for the development of a gap analysis, a self-assessment, a Gantt chart and/or an action plan. The rate of progress in developing a quality system will depend on prevailing circumstances in each country and the availability of resources. BEs should keep their long-term goal in mind and work steadily towards full implementation of a quality system and the benefits it brings.

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